

BFS Traversal

Step 1: Start at E

Complete

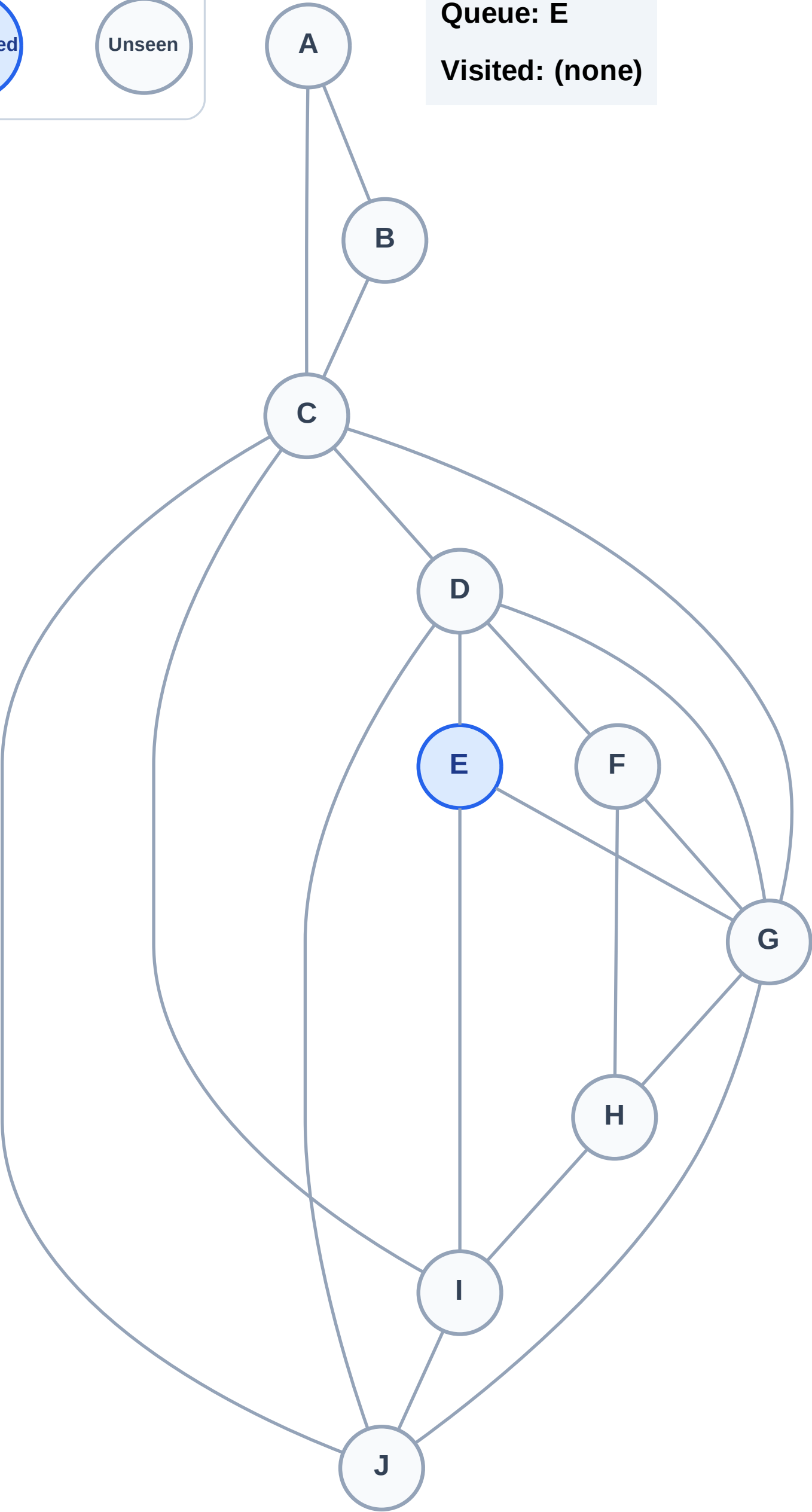
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

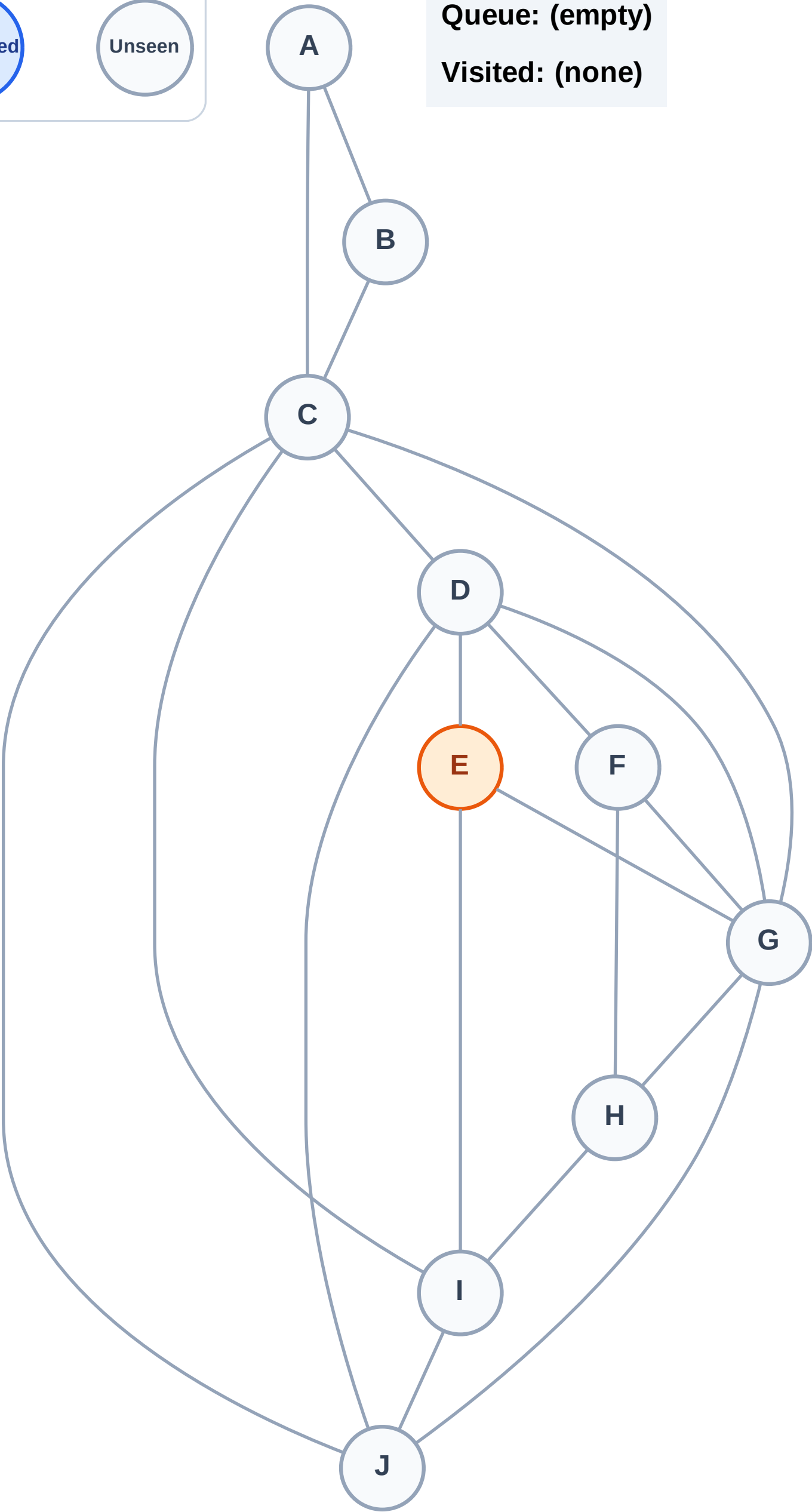
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover D, G, I from E

Legend

Complete

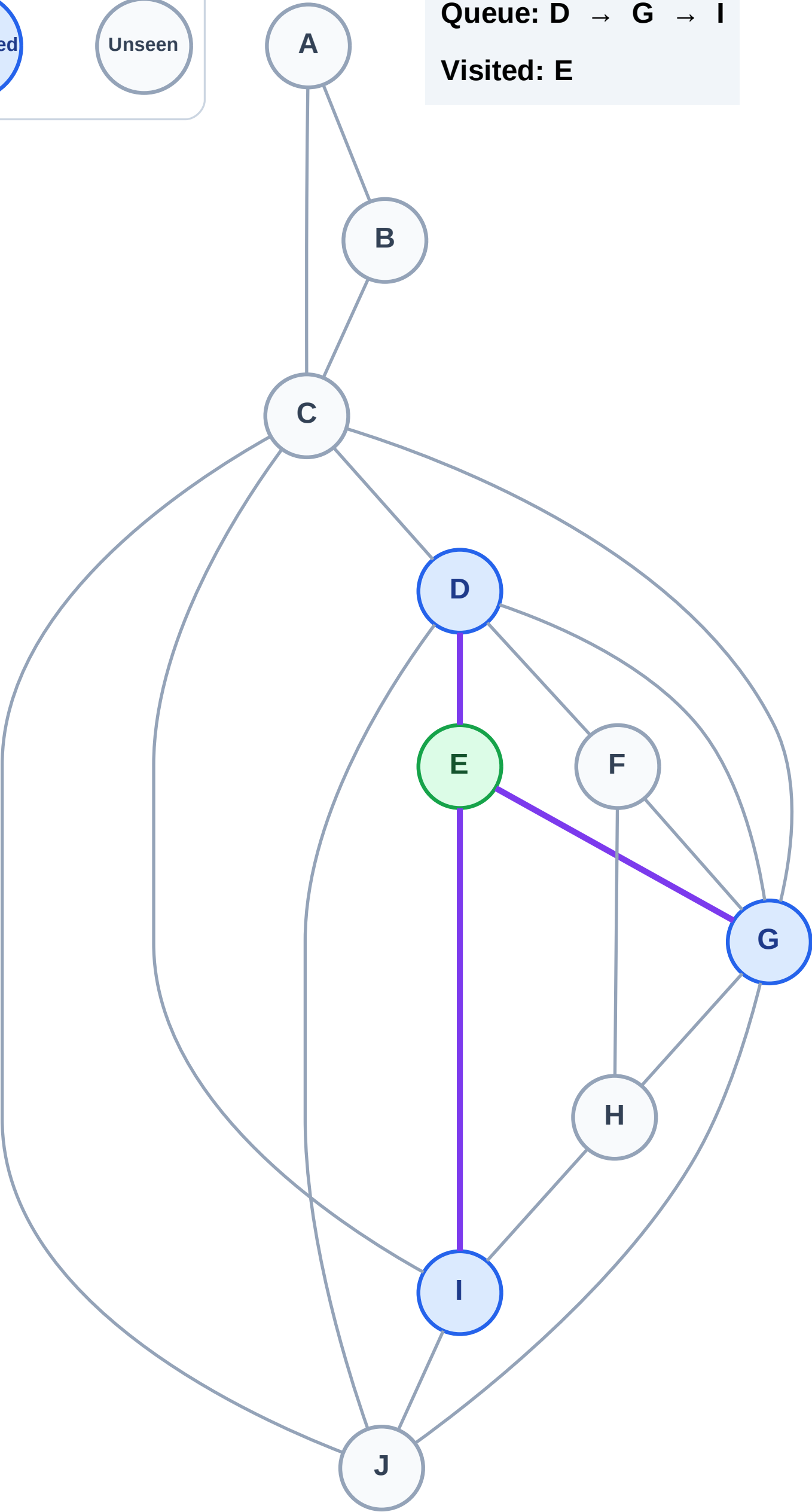
Processing

Discovered

Unseen

Queue: D → G → I

Visited: E



BFS Traversal

Step 4: Process node D

Legend

Complete

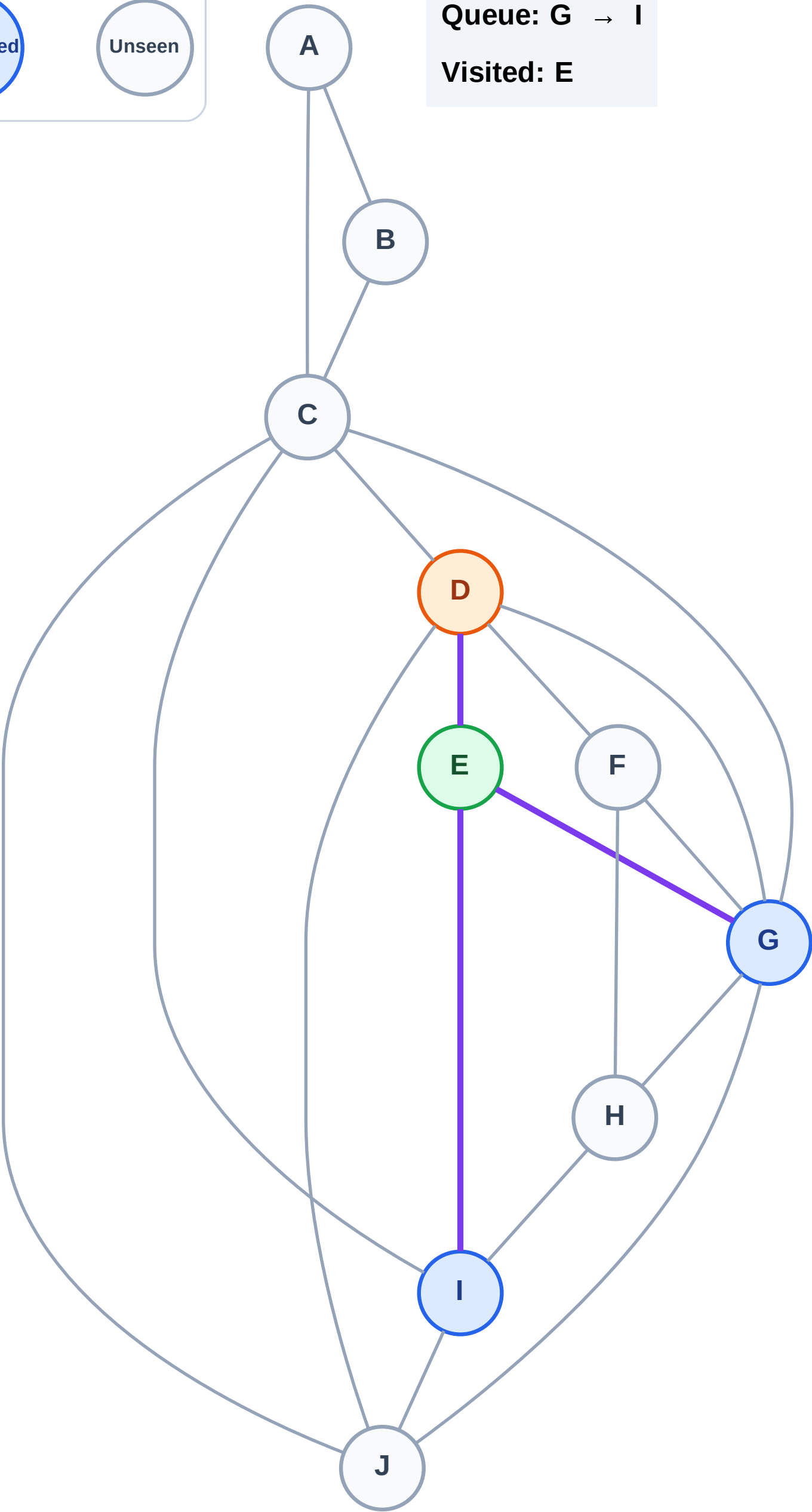
Processing

Discovered

Unseen

Queue: G → I

Visited: E



BFS Traversal

Step 5: Discover C, F, J from D

Legend

Complete

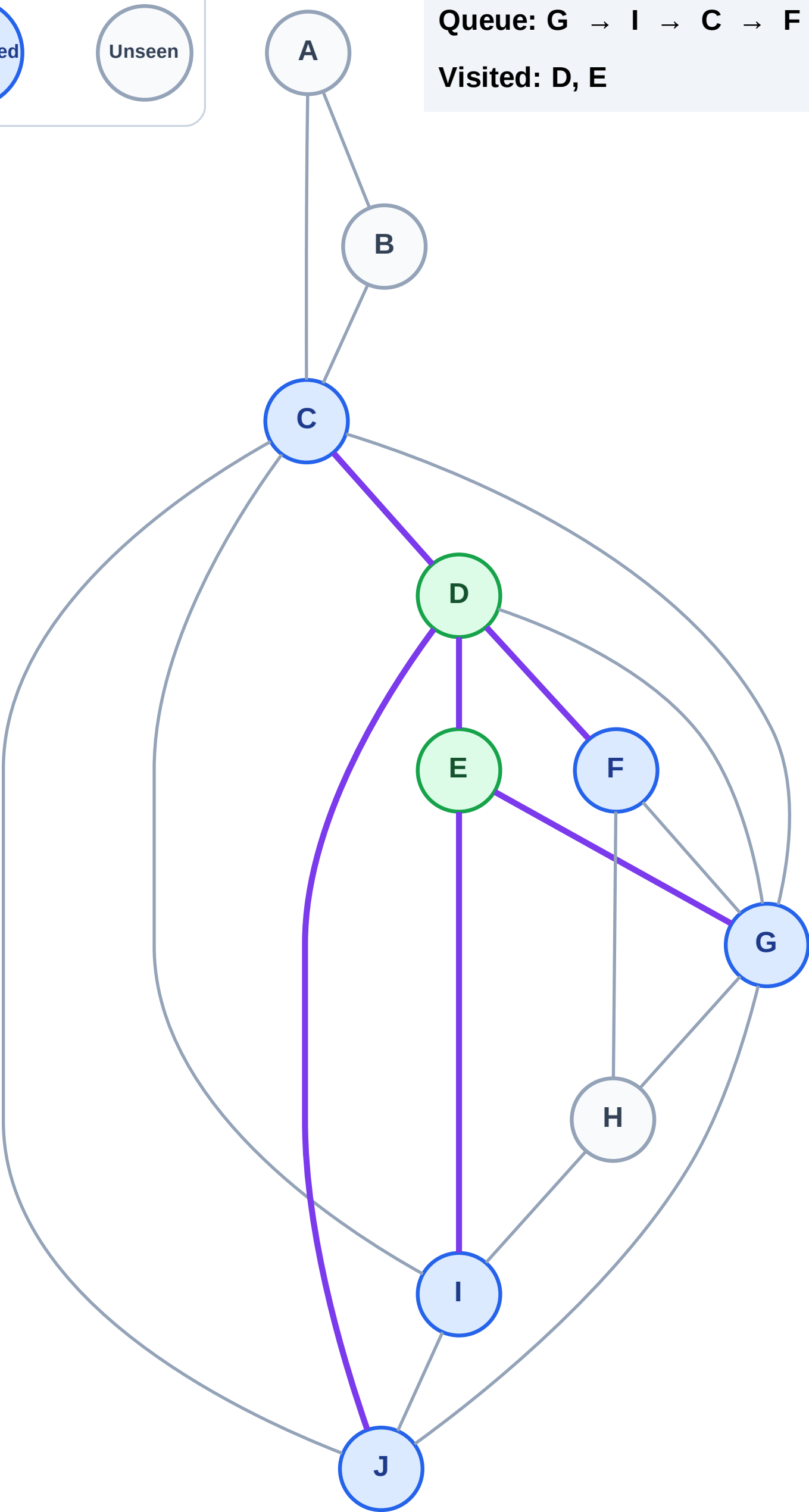
Processing

Discovered

Unseen

Queue: G → I → C → F → J

Visited: D, E



BFS Traversal

Step 6: Process node G

Legend

Complete

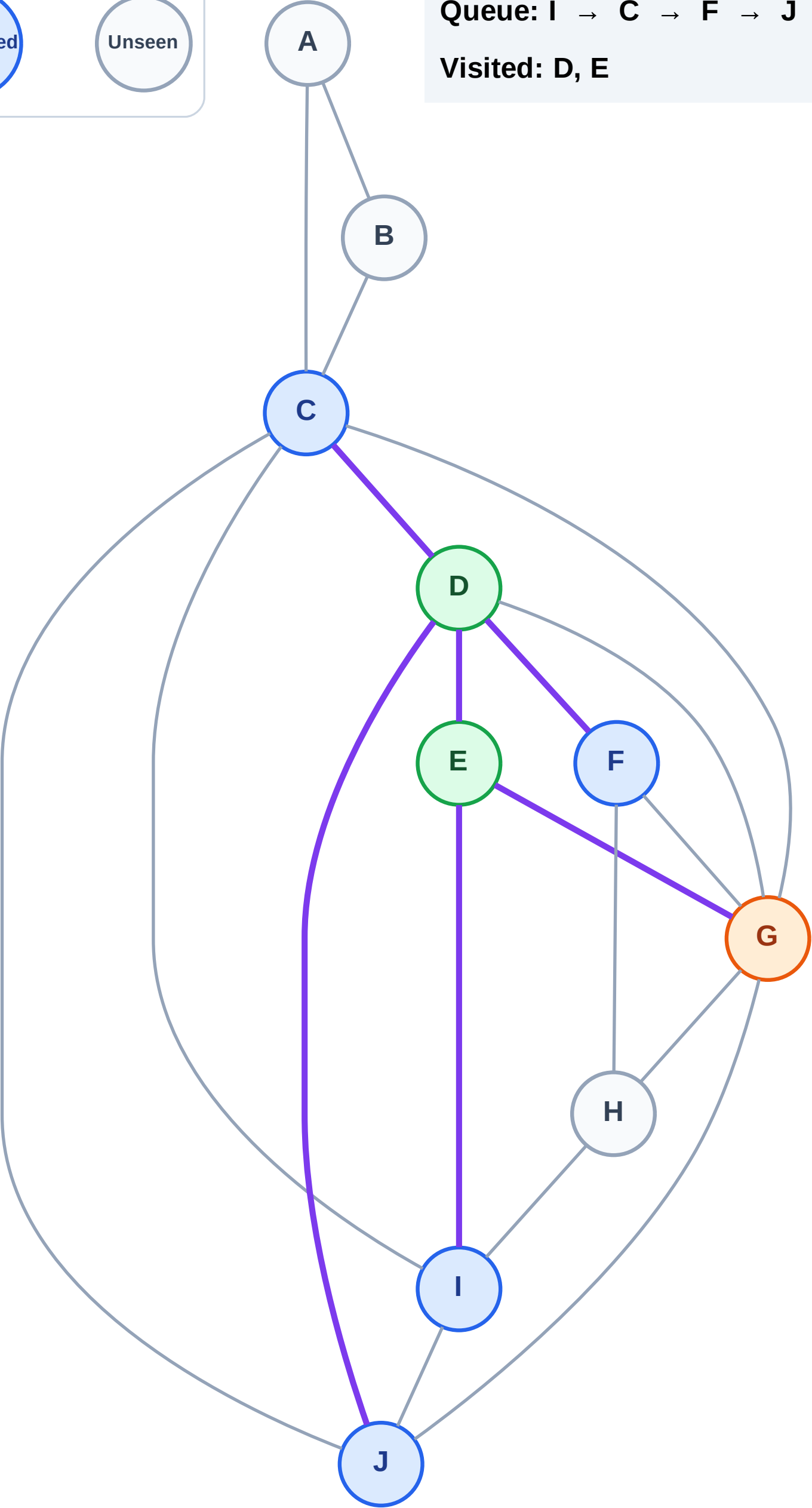
Processing

Discovered

Unseen

Queue: I → C → F → J

Visited: D, E



BFS Traversal

Step 7: Discover H from G

Legend

Complete

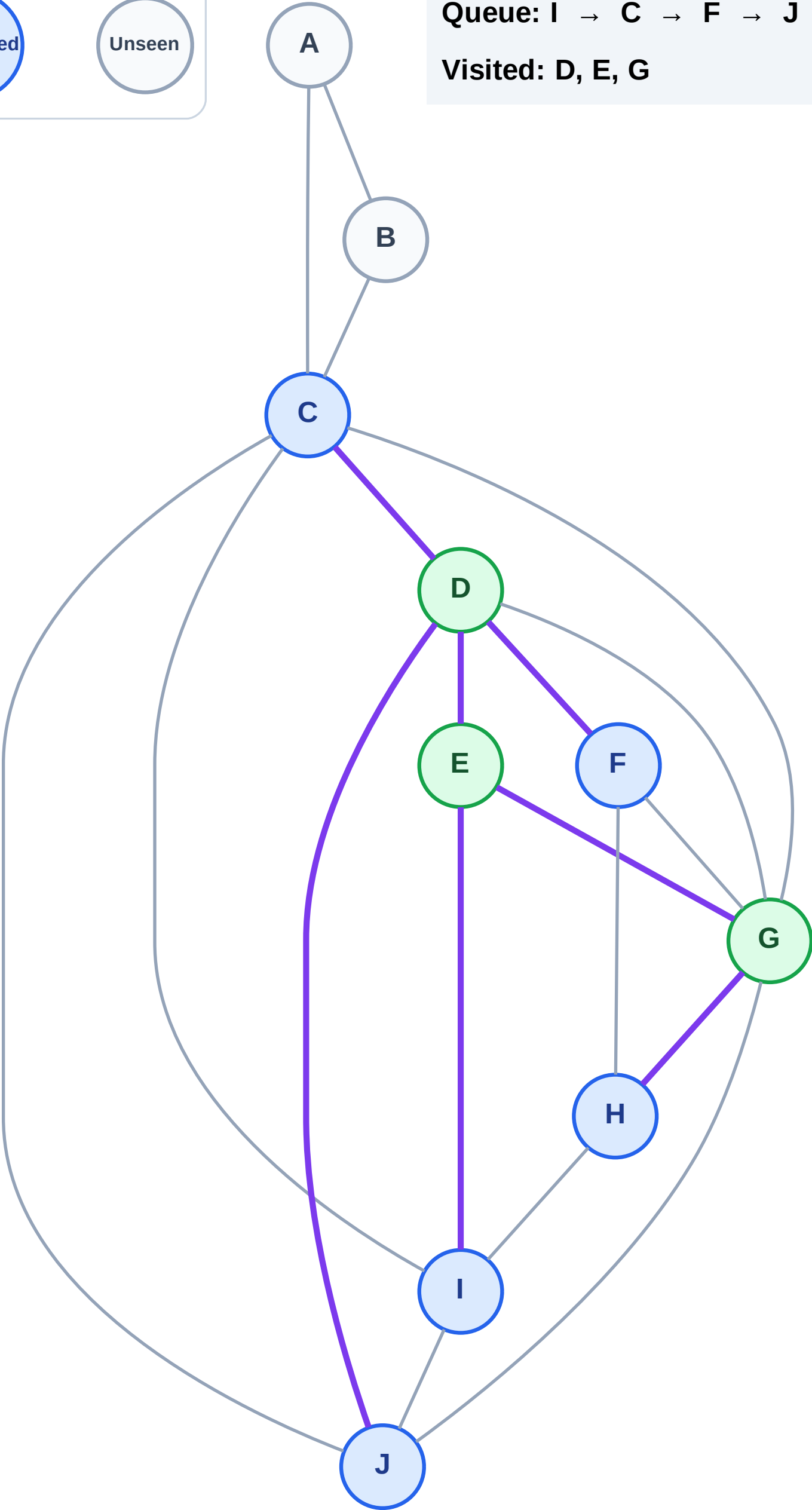
Processing

Discovered

Unseen

Queue: I → C → F → J → H

Visited: D, E, G



BFS Traversal

Step 8: Process node I

Legend

Complete

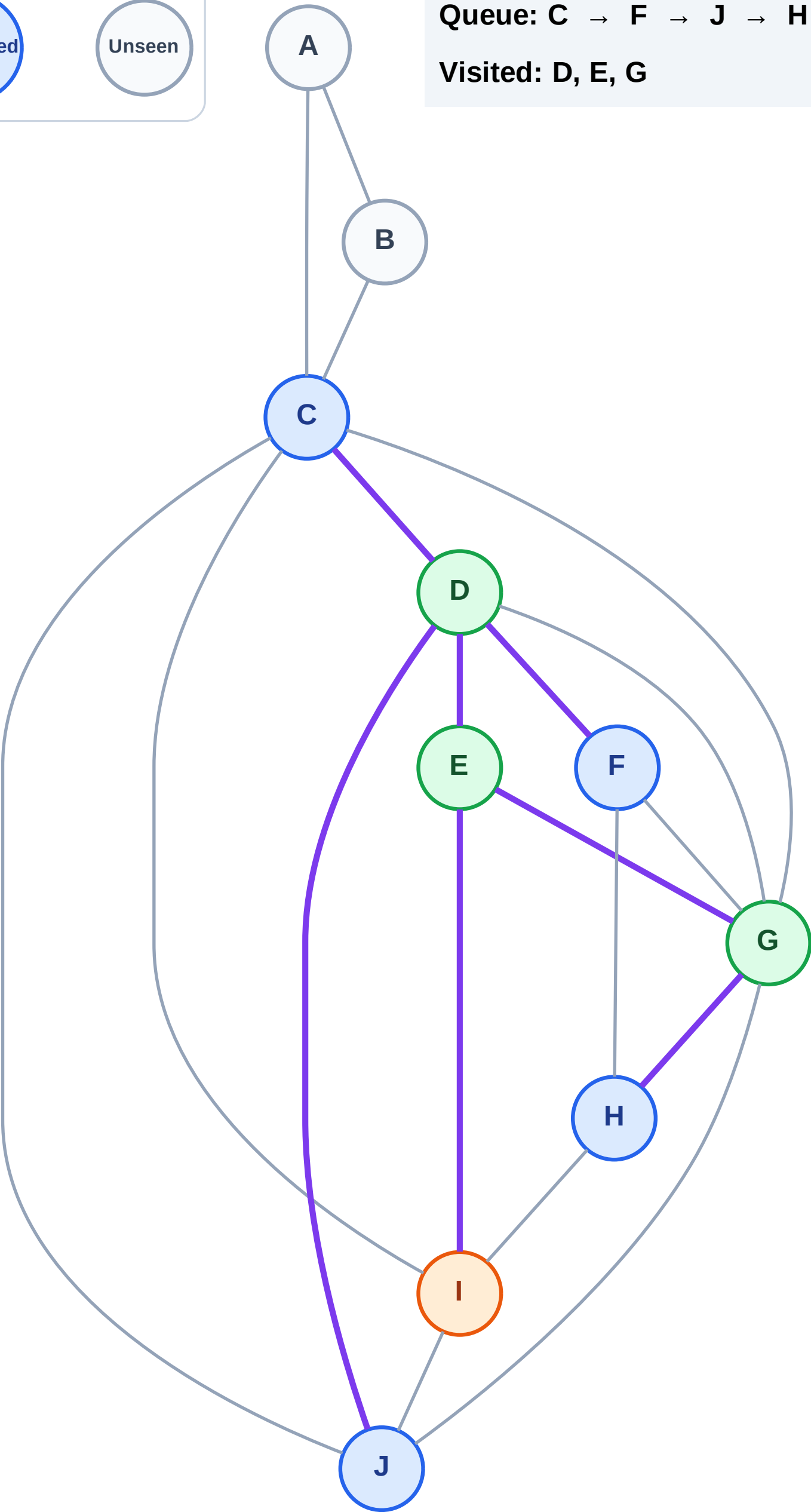
Processing

Discovered

Unseen

Queue: C → F → J → H

Visited: D, E, G



BFS Traversal

Step 9: No new neighbors from I

Legend

Complete

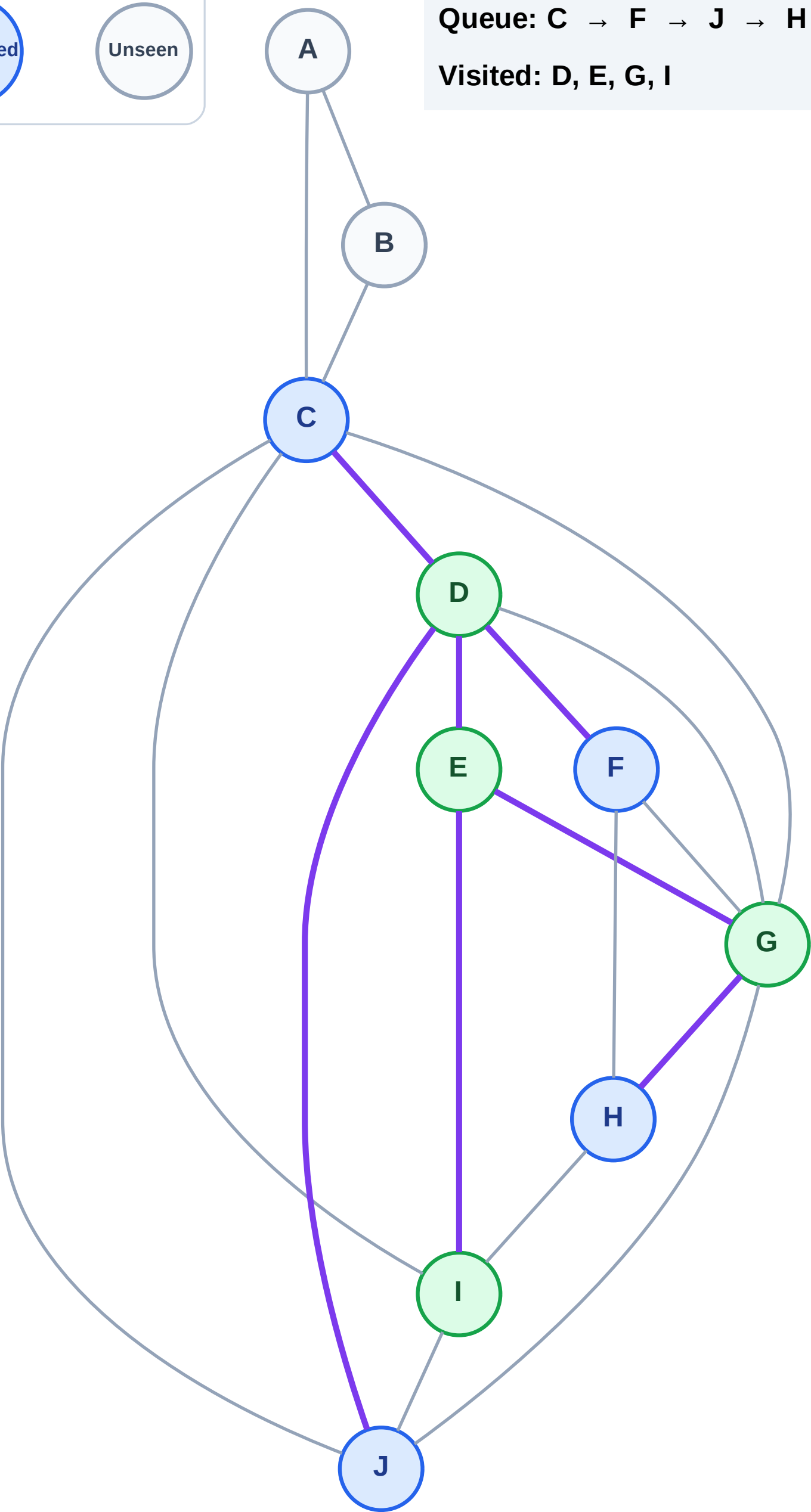
Processing

Discovered

Unseen

Queue: C → F → J → H

Visited: D, E, G, I



BFS Traversal

Step 10: Process node C

Legend

Complete

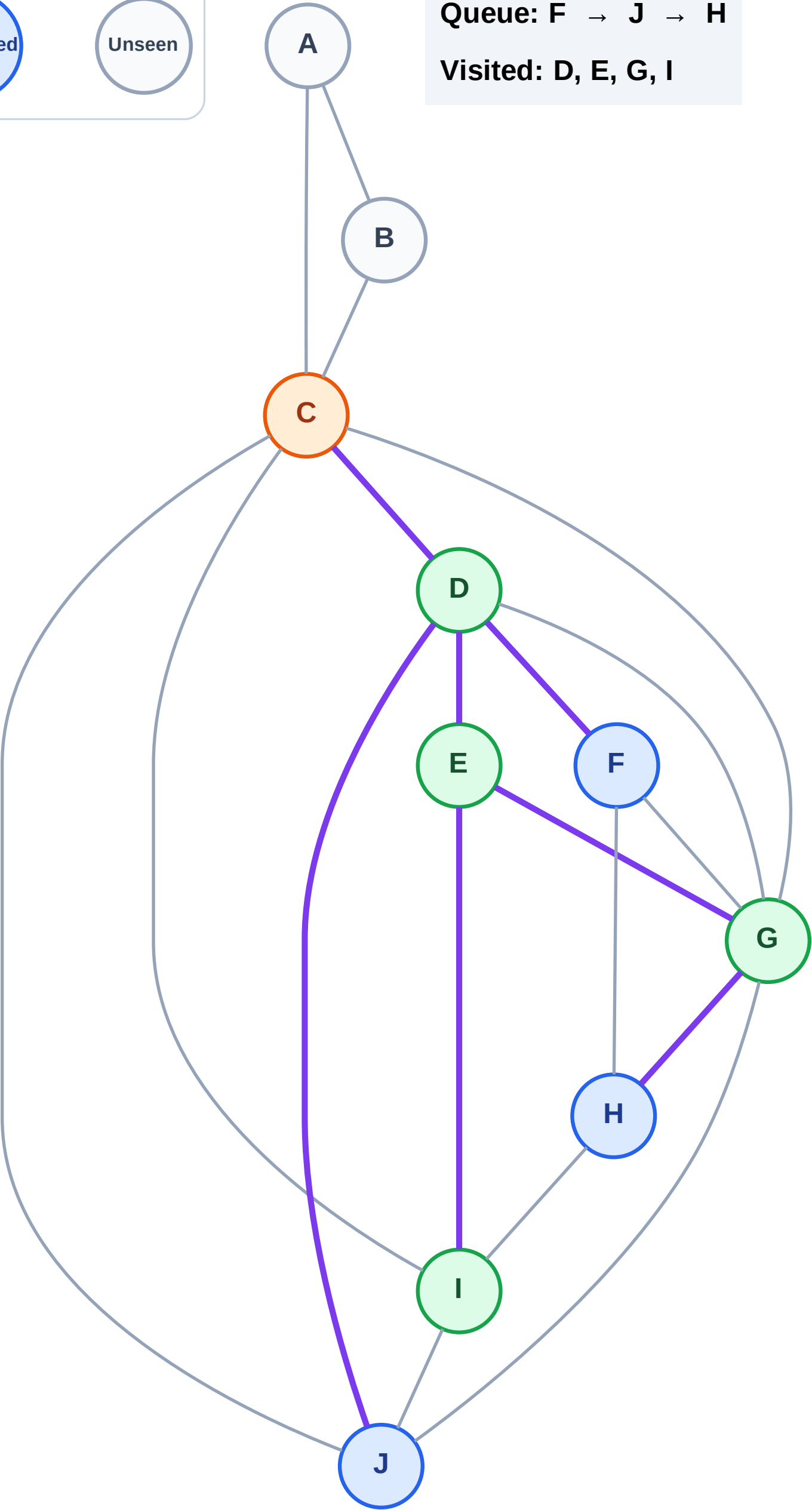
Processing

Discovered

Unseen

Queue: F → J → H

Visited: D, E, G, I



BFS Traversal

Step 11: Discover A, B from C

Legend



Complete



Processing



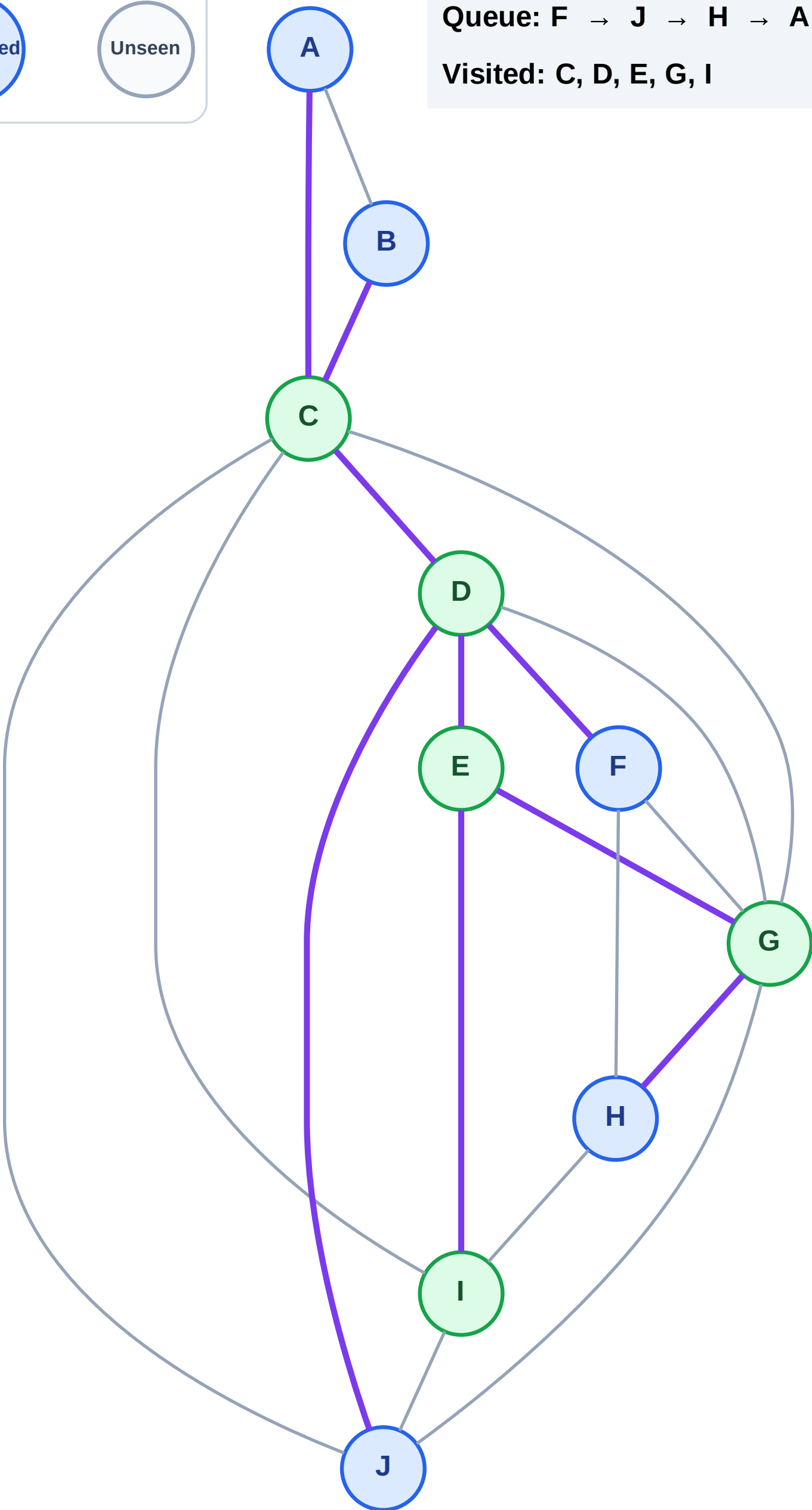
Discovered



Unseen

Queue: F → J → H → A → B

Visited: C, D, E, G, I



BFS Traversal

Step 12: Process node F

Legend

Complete

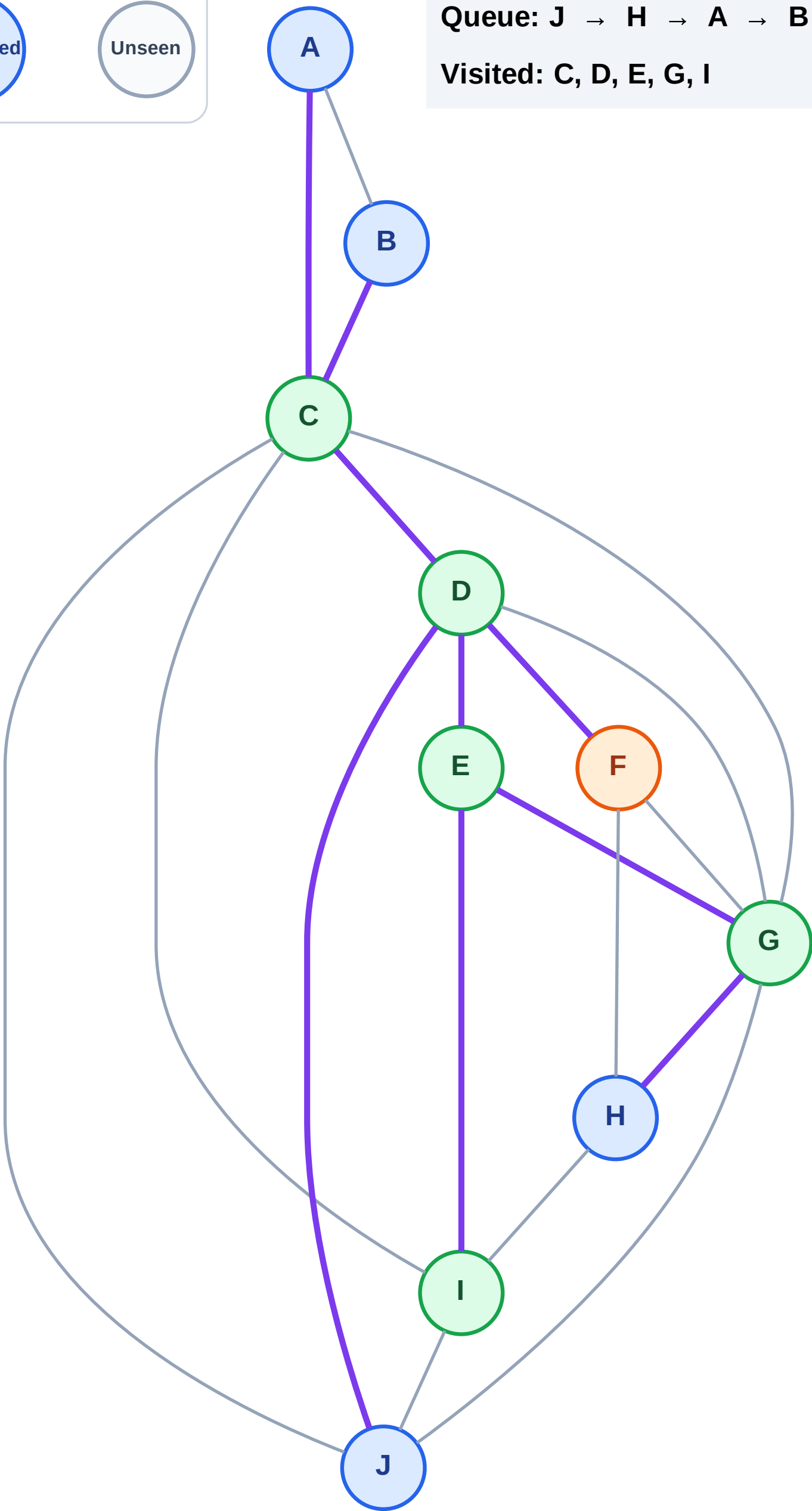
Processing

Discovered

Unseen

Queue: J → H → A → B

Visited: C, D, E, G, I



BFS Traversal

Step 13: No new neighbors from F

Complete

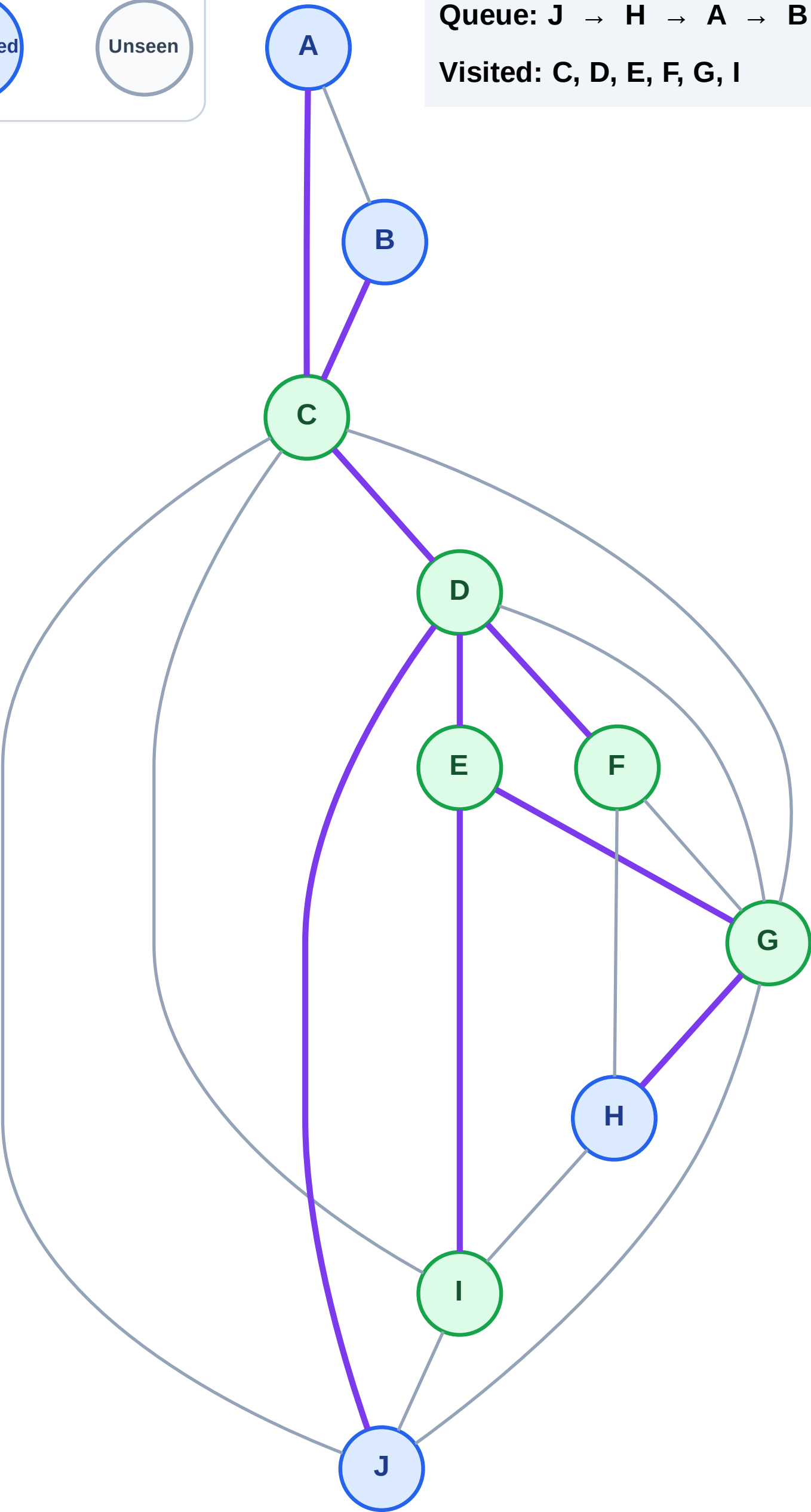
Processing

Discovered

Unseen

Queue: J → H → A → B

Visited: C, D, E, F, G, I



BFS Traversal

Step 14: Process node J

Legend

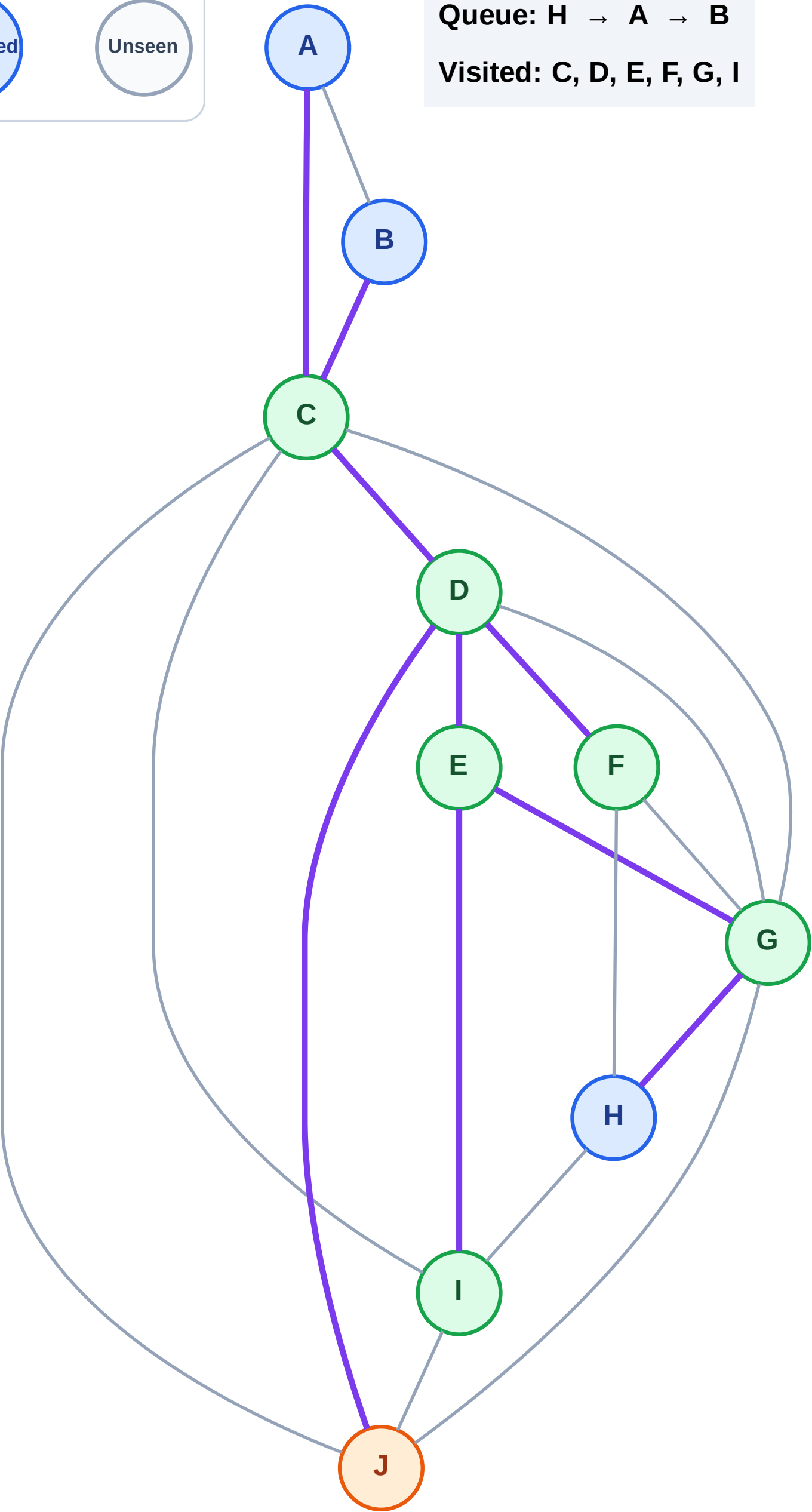
Complete

Processing

Discovered

Unseen

Queue: H → A → B
Visited: C, D, E, F, G, I



BFS Traversal

Step 15: No new neighbors from J

Complete

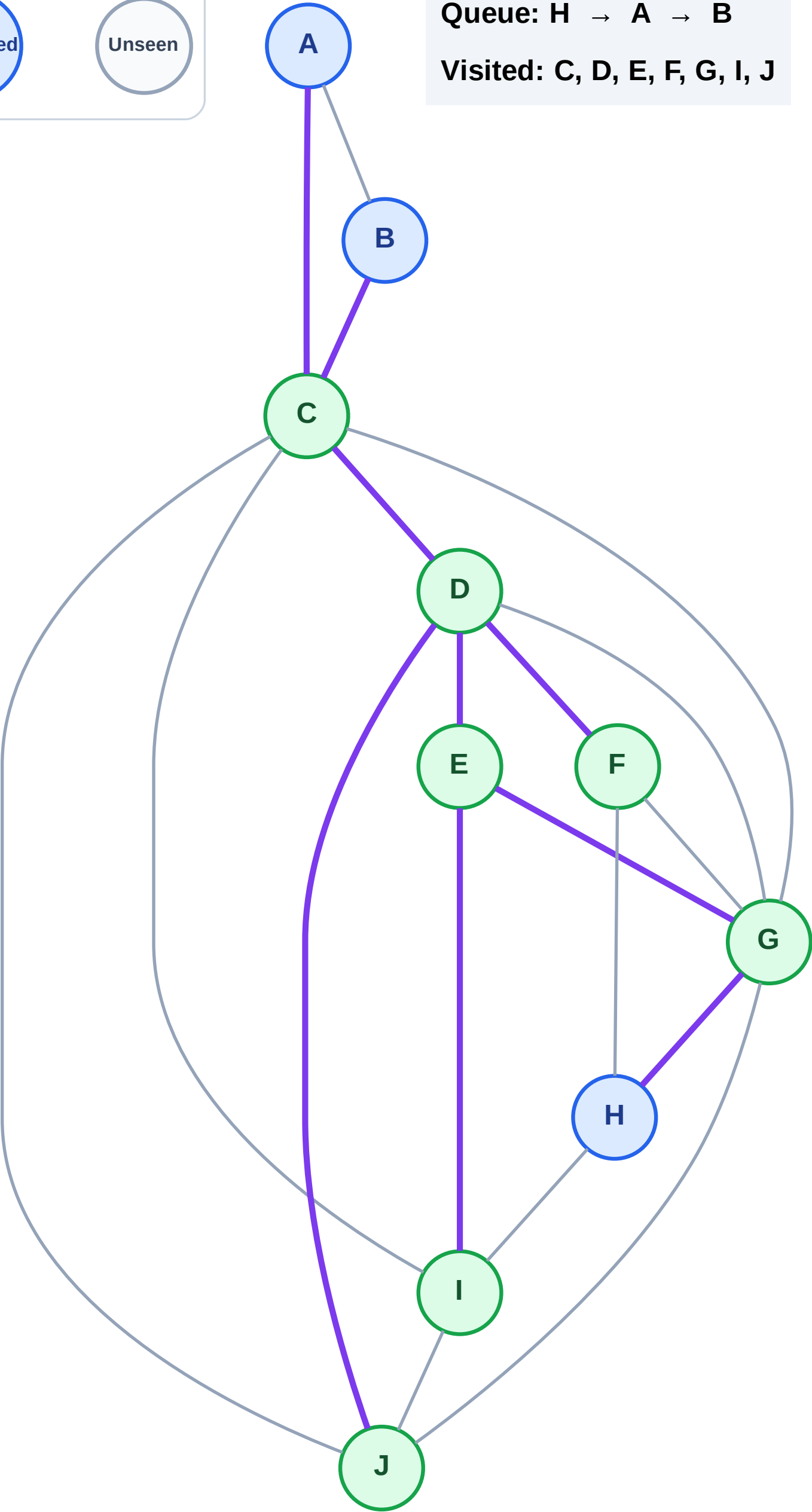
Processing

Discovered

Unseen

Queue: H → A → B

Visited: C, D, E, F, G, I, J



BFS Traversal

Step 16: Process node H

Complete

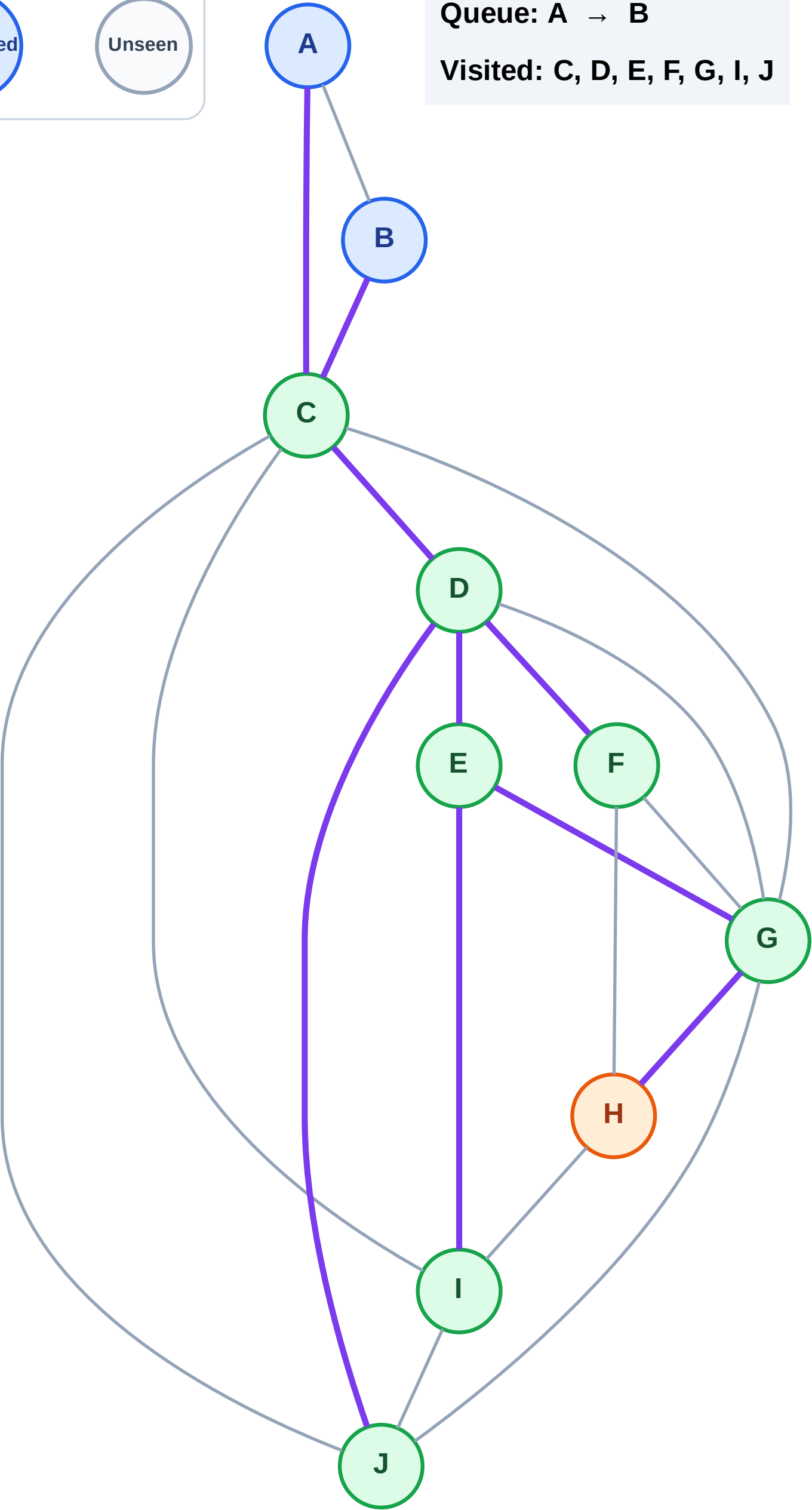
Processing

Discovered

Unseen

Queue: A → B

Visited: C, D, E, F, G, I, J



BFS Traversal

Step 17: No new neighbors from H

Legend

Complete

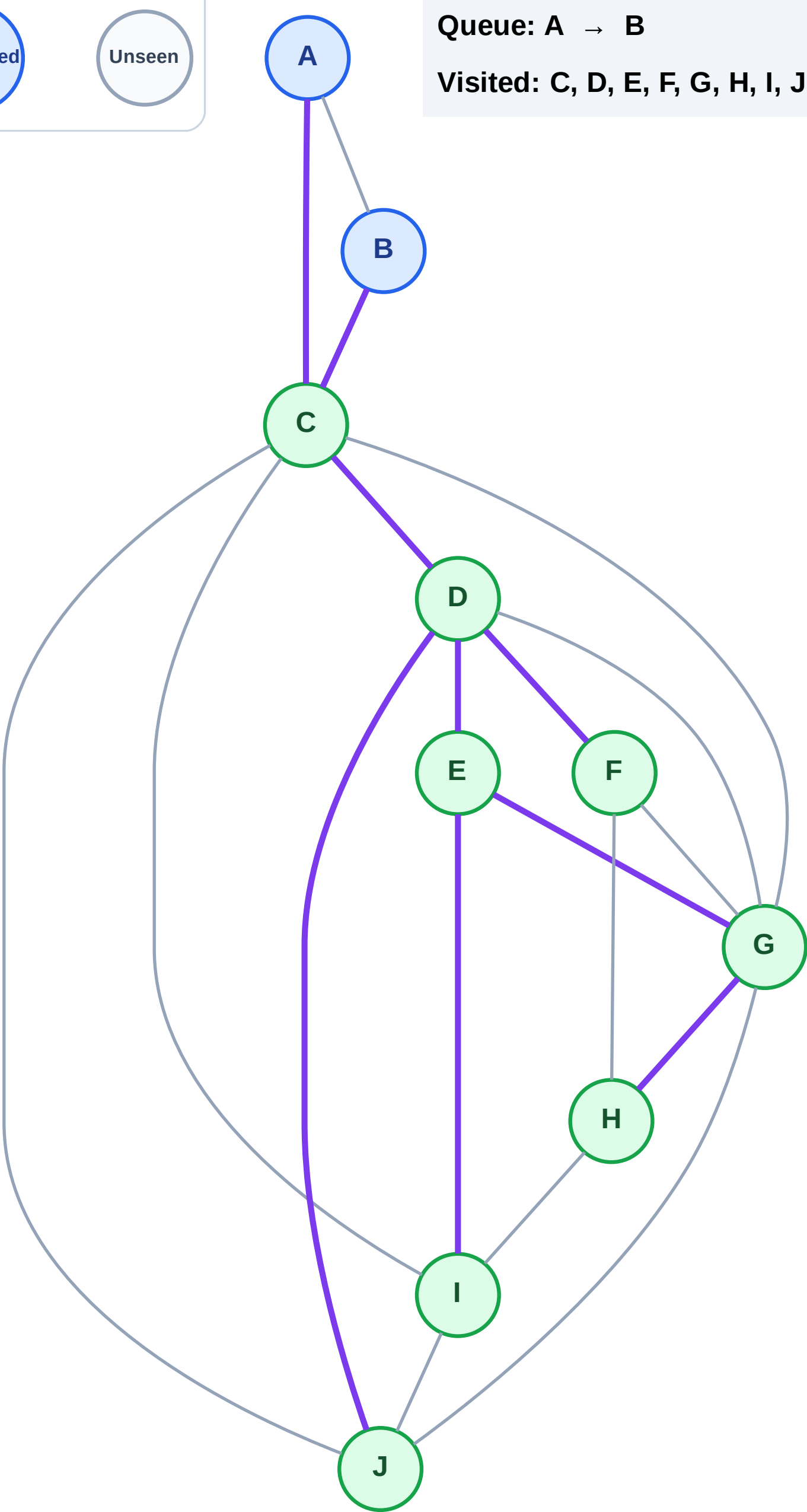
Processing

Discovered

Unseen

Queue: A → B

Visited: C, D, E, F, G, H, I, J



BFS Traversal

Step 18: Process node A

Legend

Complete

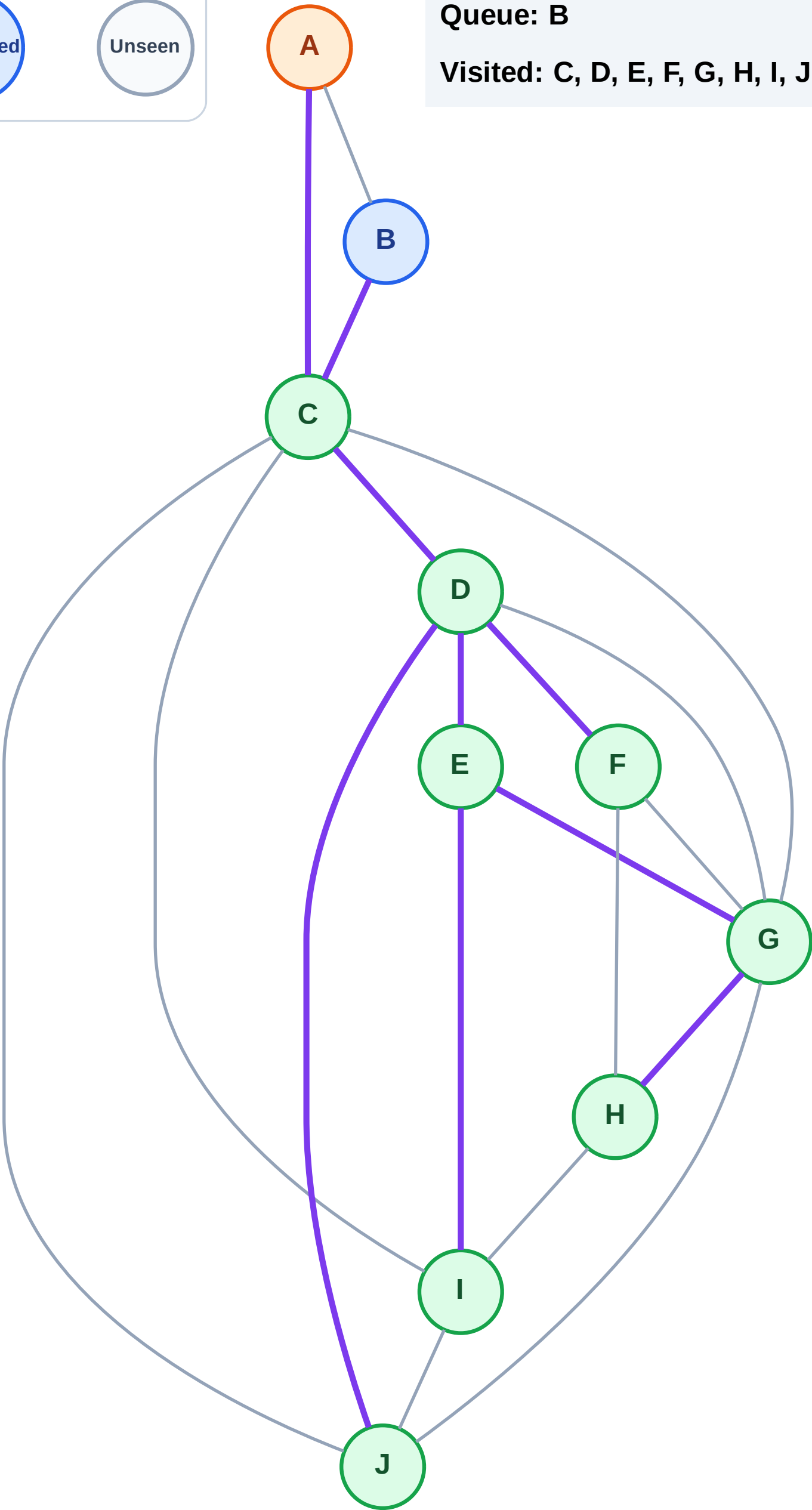
Processing

Discovered

Unseen

Queue: B

Visited: C, D, E, F, G, H, I, J



BFS Traversal

Step 19: No new neighbors from A

Legend

Complete

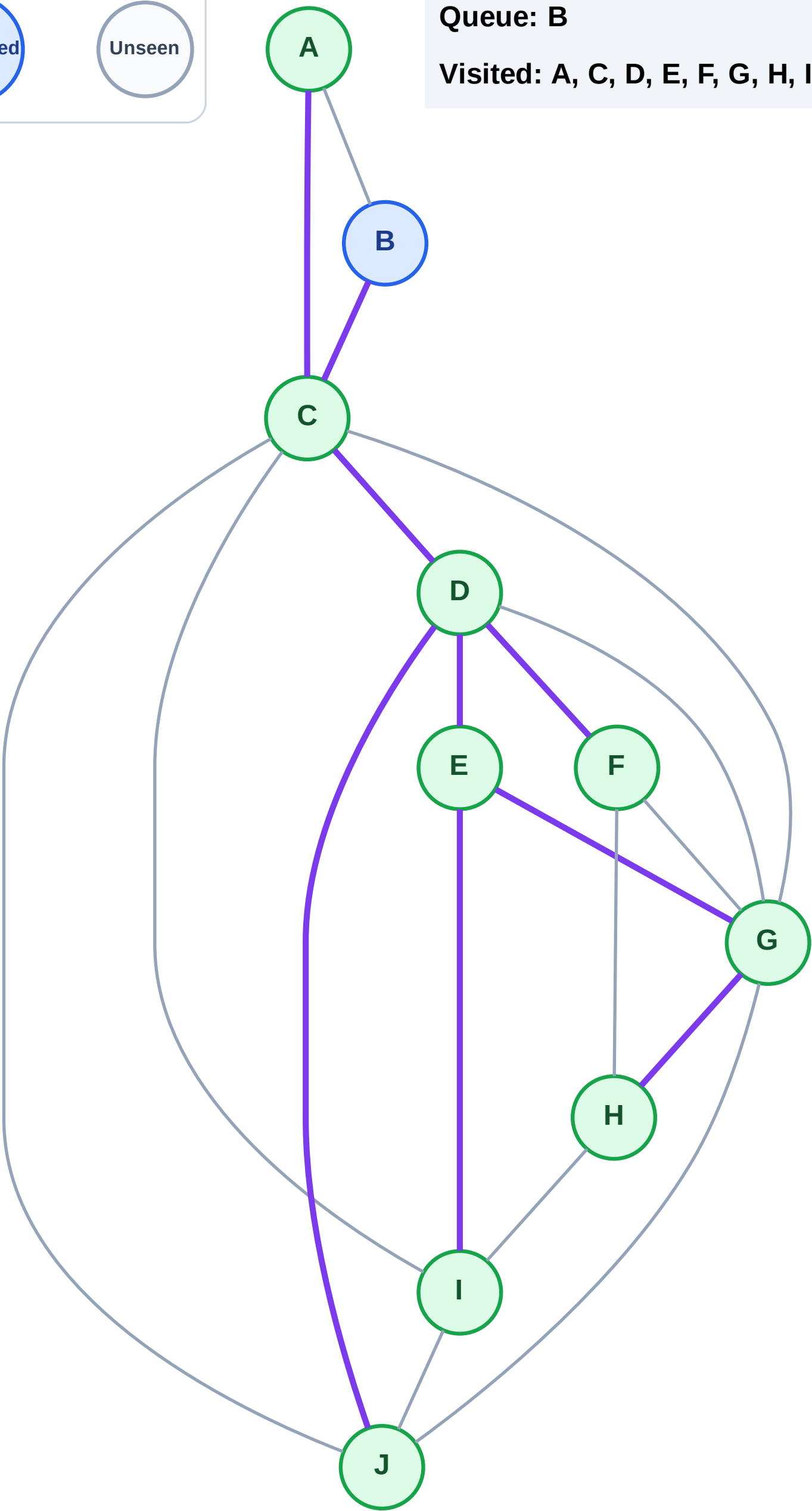
Processing

Discovered

Unseen

Queue: B

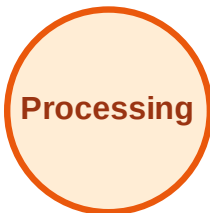
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

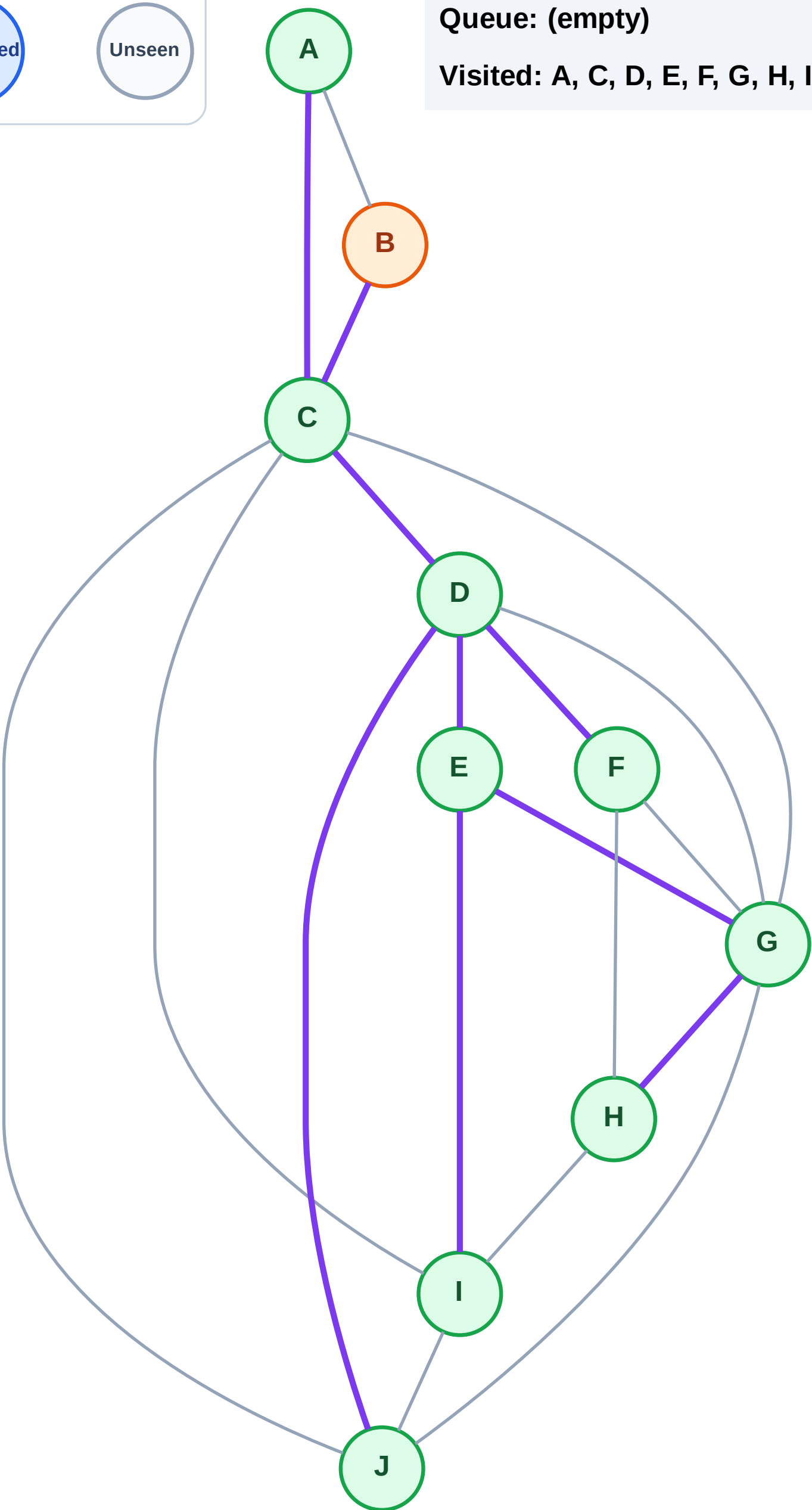
Step 20: Process node B

Legend



Queue: (empty)

Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

Step 21: No new neighbors from B

Legend

Complete

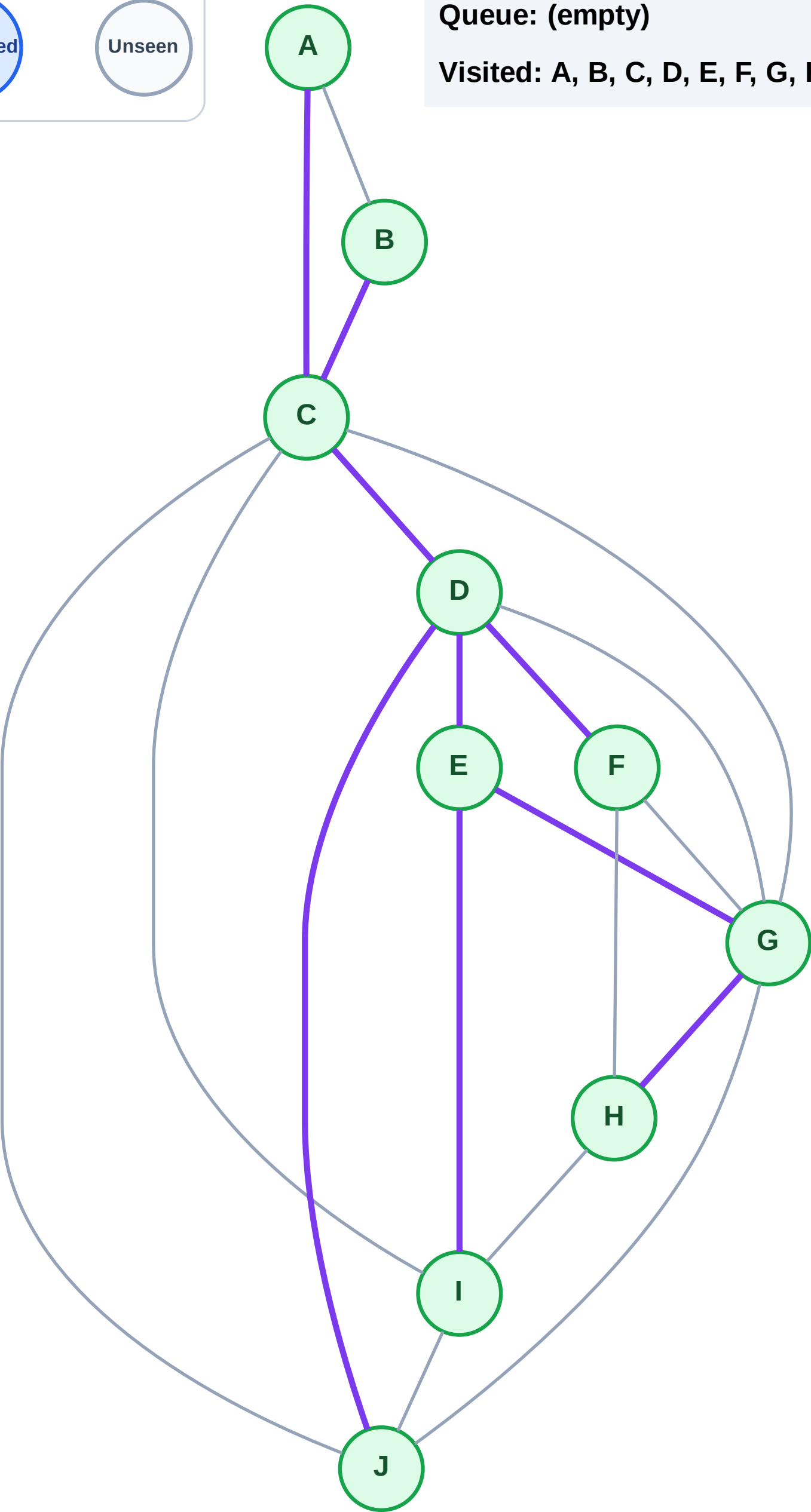
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

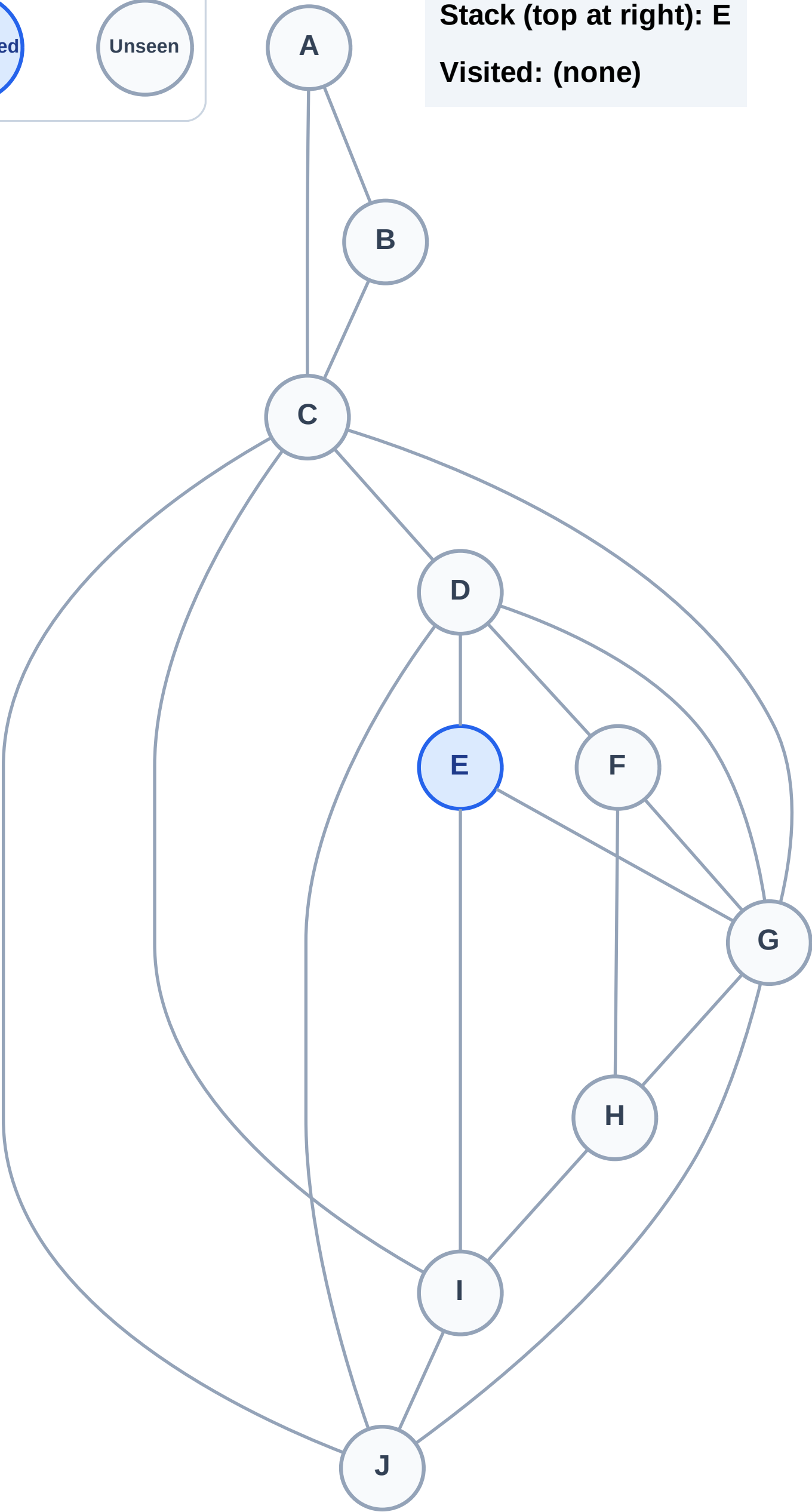
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

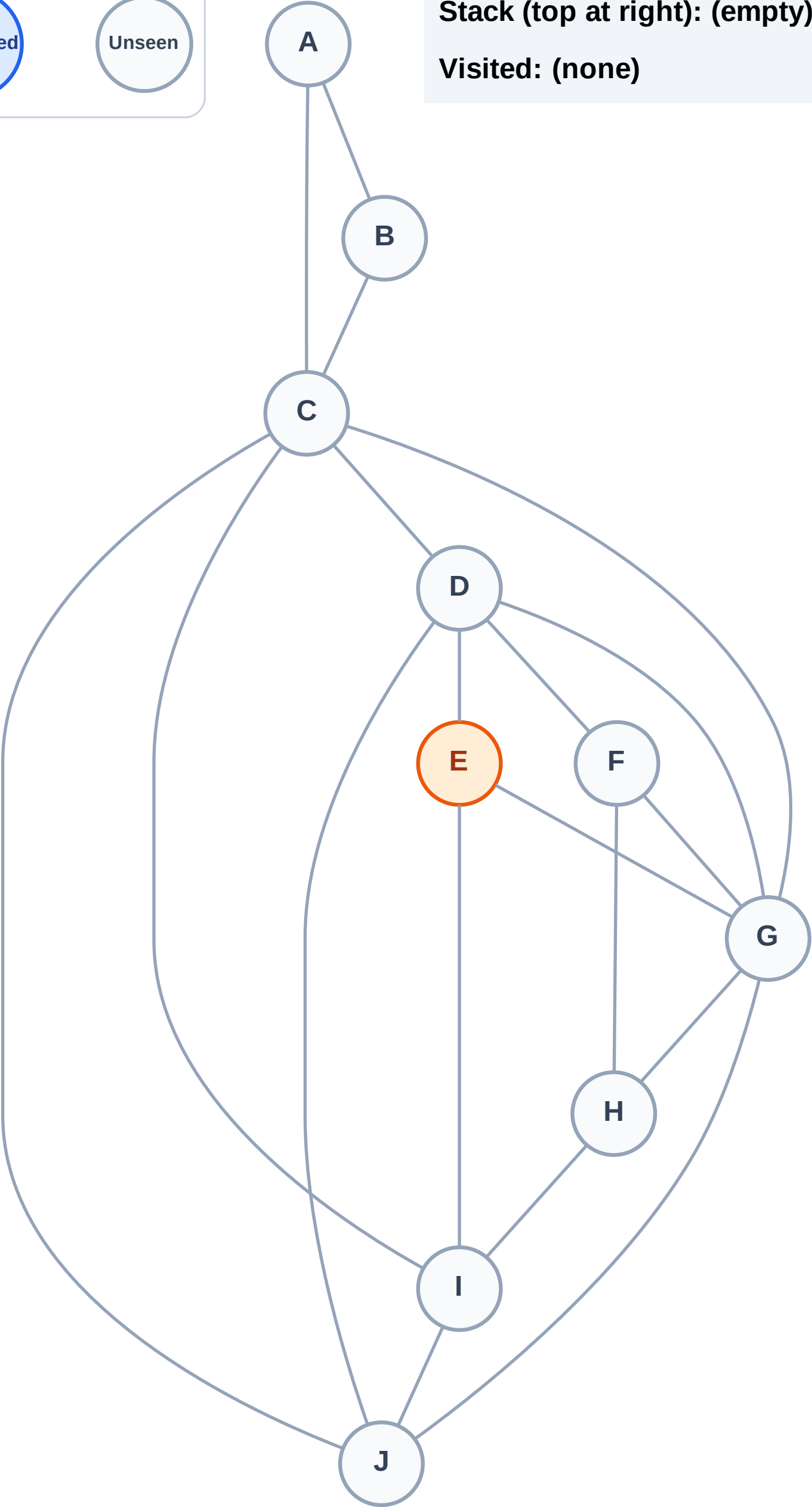
Complete

Processing

Discovered

Unseen

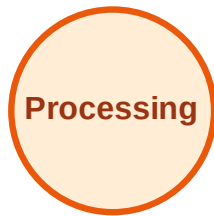
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

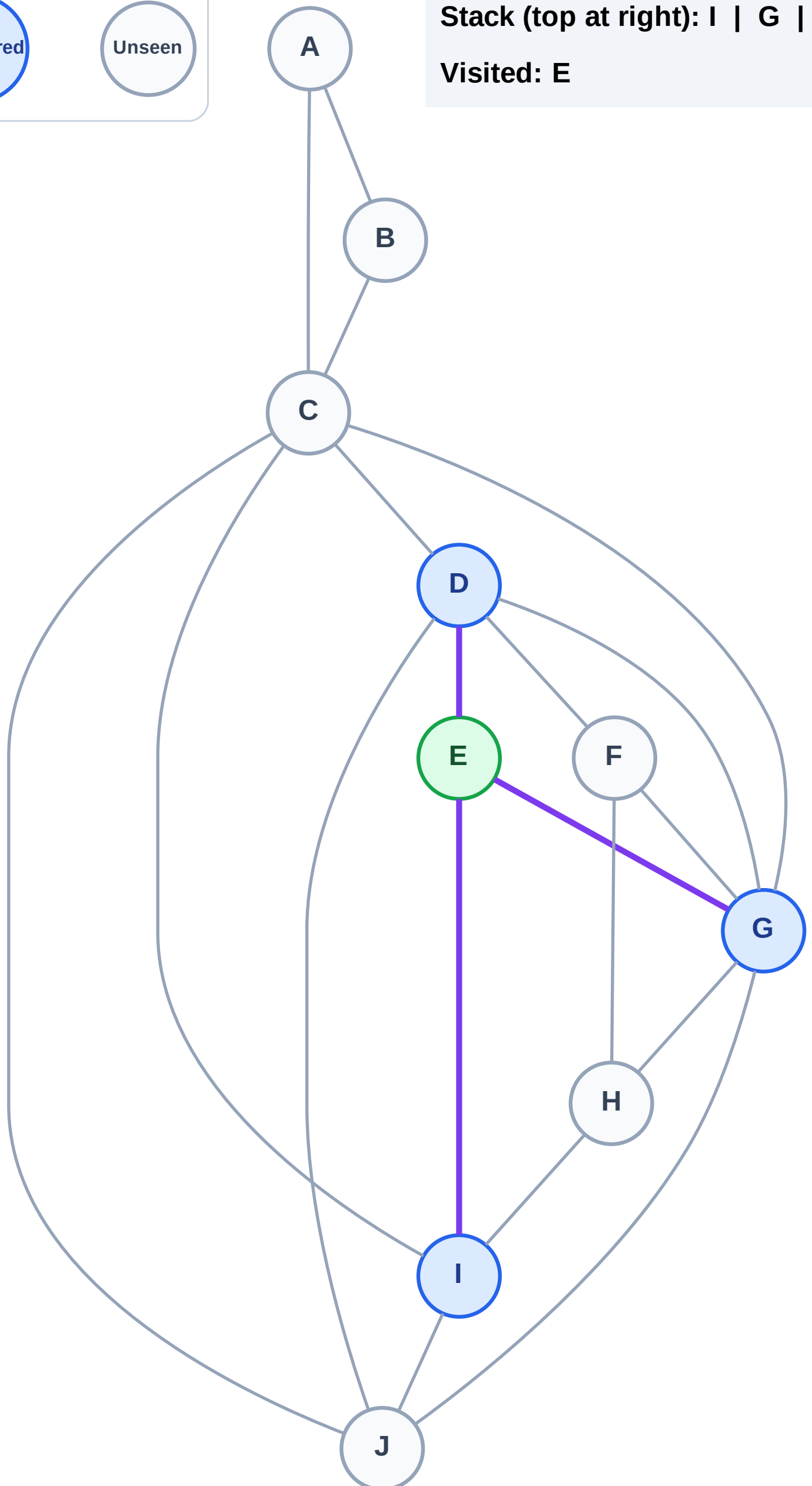
Step 3: Discover I, G, D from E

Legend



Stack (top at right): I | G | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

Complete

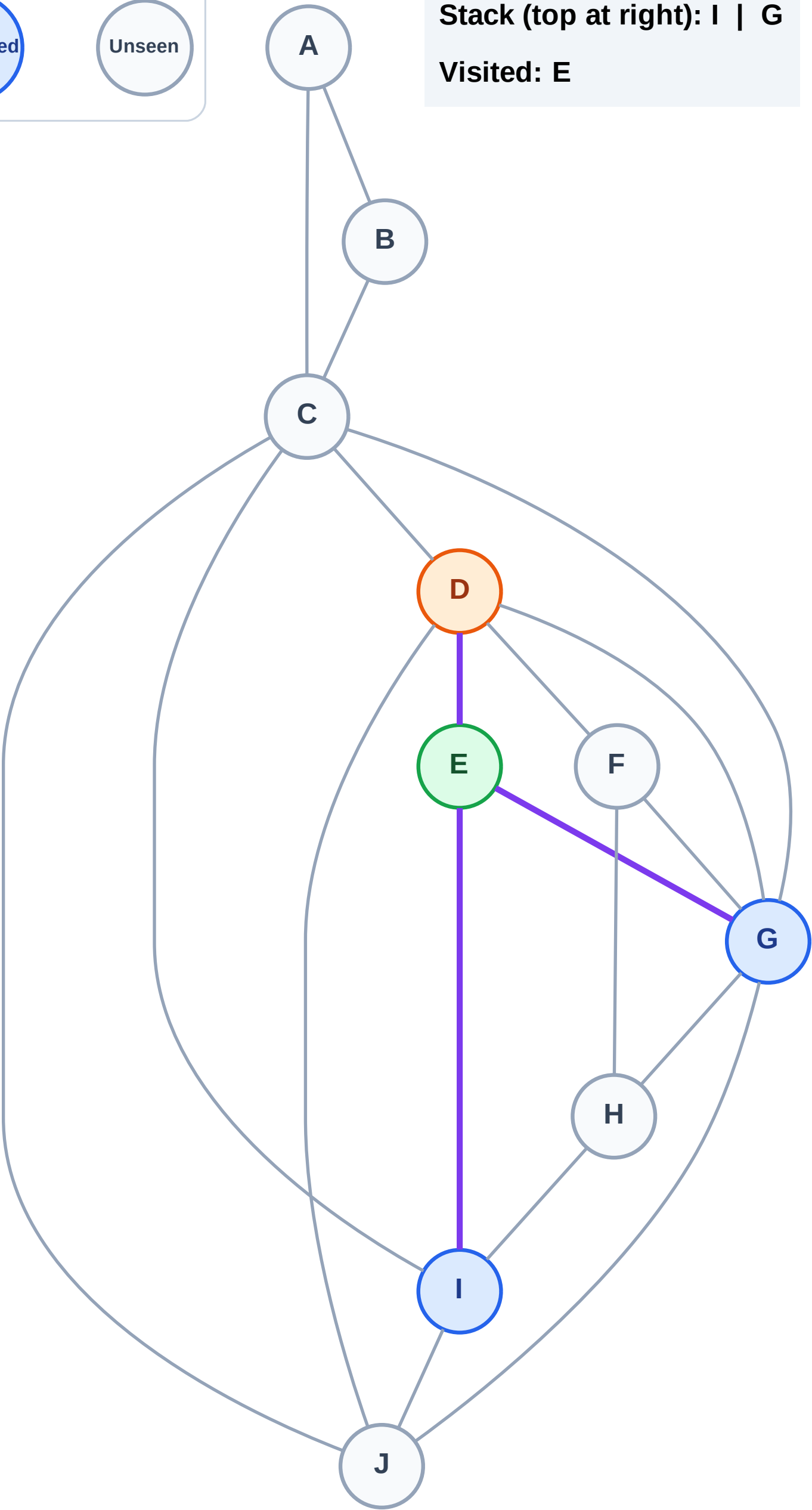
Processing

Discovered

Unseen

Stack (top at right): I | G

Visited: E



DFS Traversal

Step 5: Discover J, F, C from D

Legend

Complete

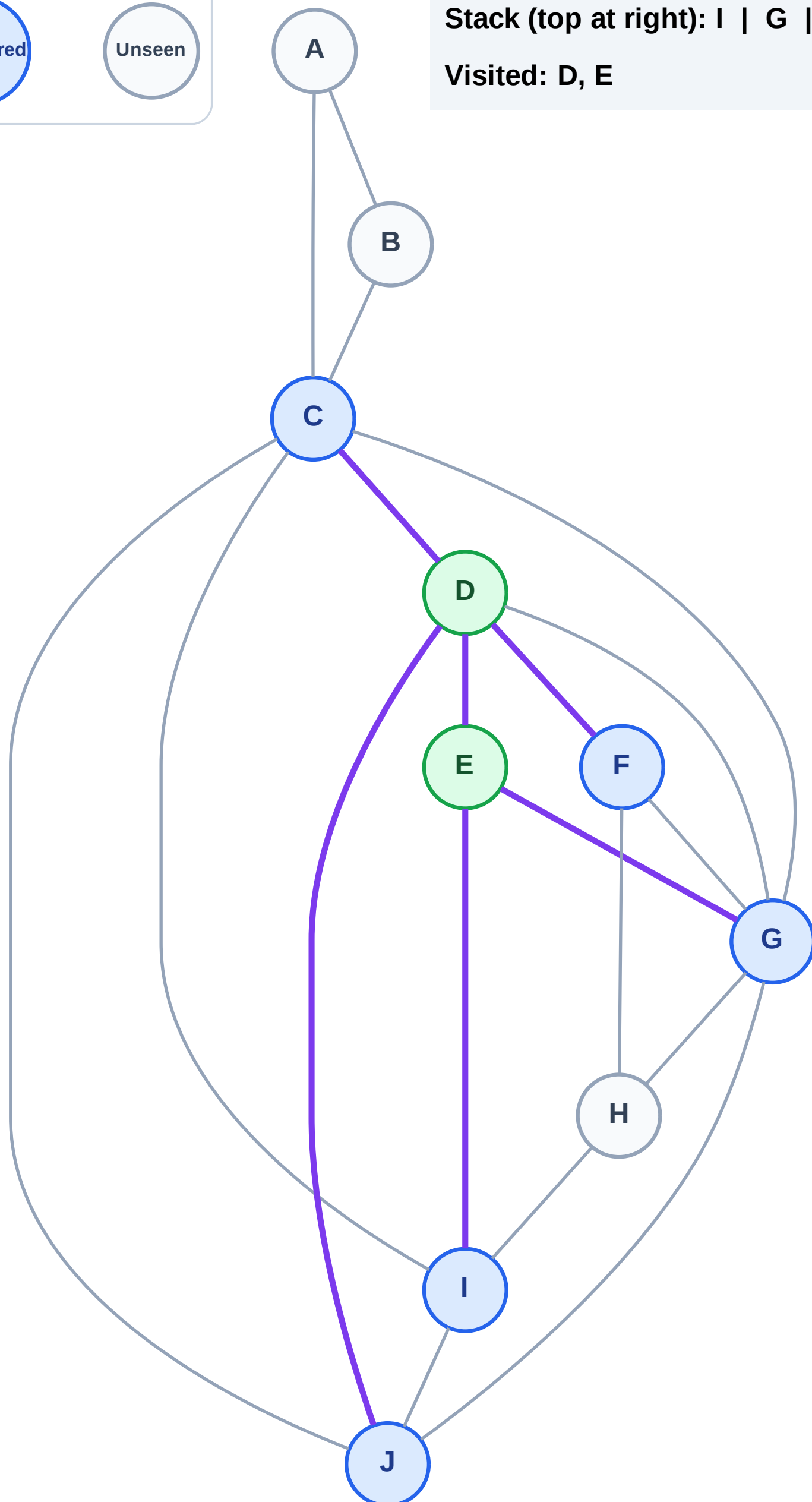
Processing

Discovered

Unseen

Stack (top at right): I | G | J | F | C

Visited: D, E



DFS Traversal

Step 6: Process node C

Legend

Complete

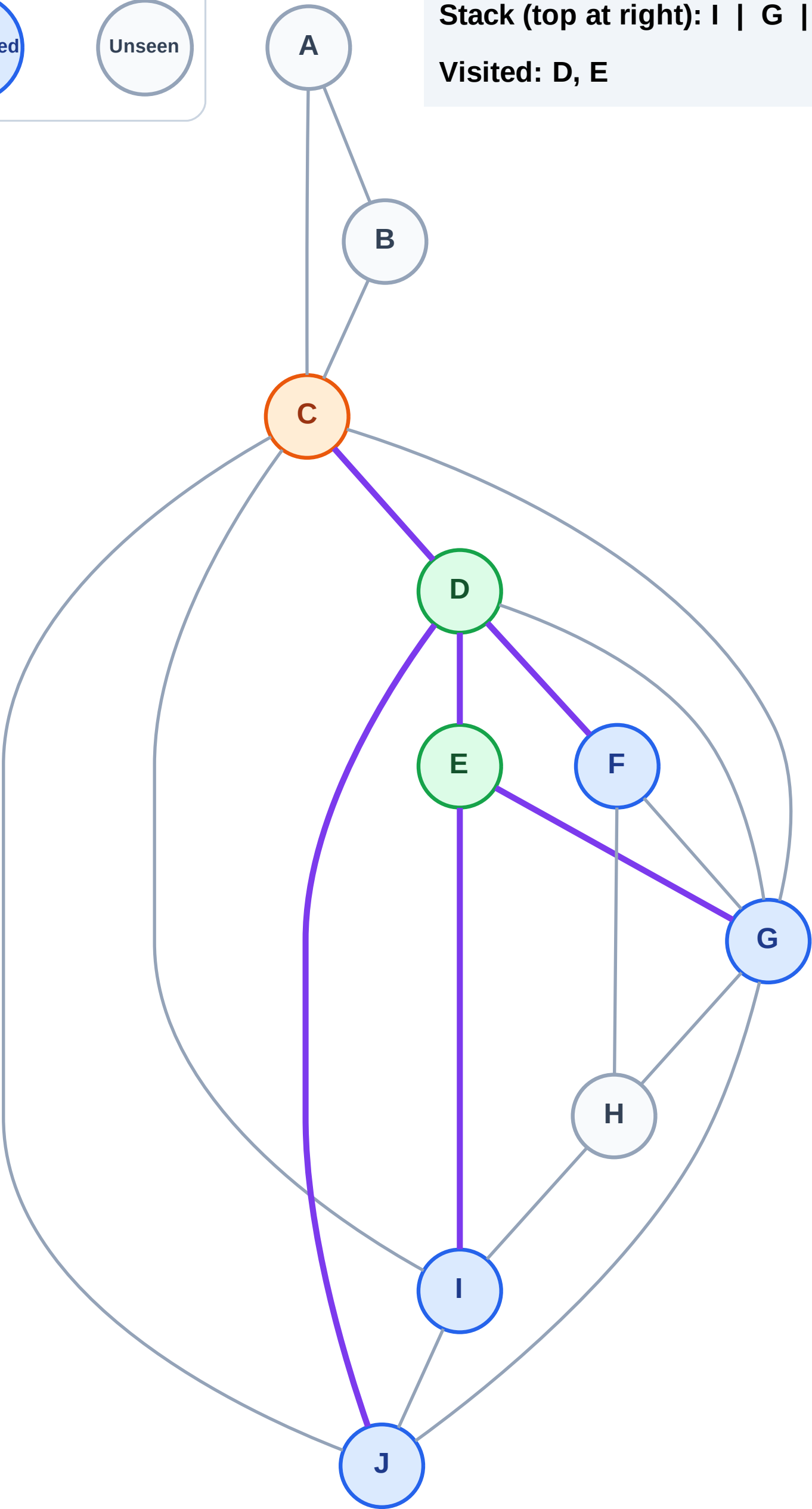
Processing

Discovered

Unseen

Stack (top at right): I | G | J | F

Visited: D, E



DFS Traversal

Step 7: Discover B, A from C

Legend

Complete

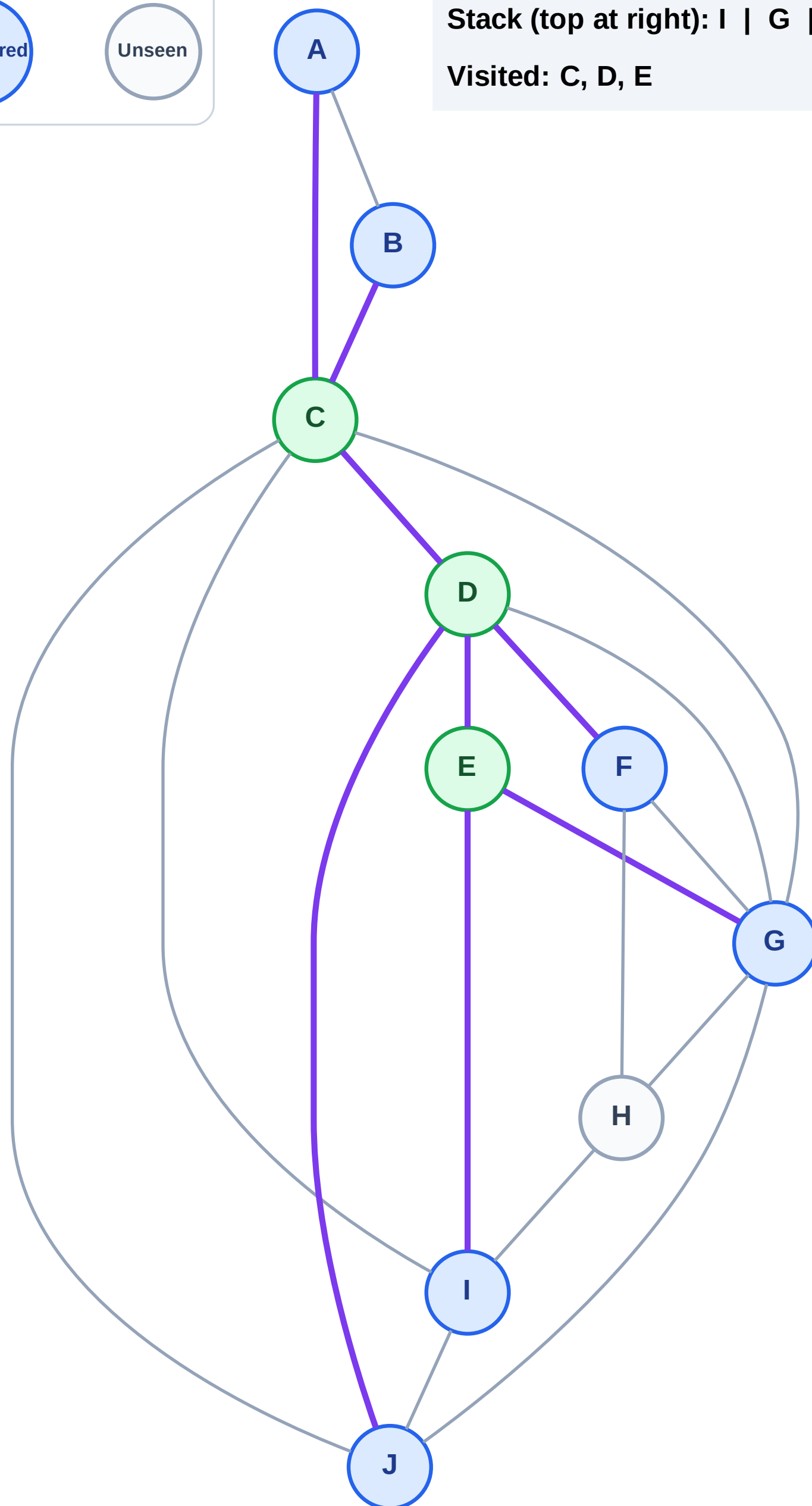
Processing

Discovered

Unseen

Stack (top at right): I | G | J | F | B | A

Visited: C, D, E



DFS Traversal

Step 8: Process node A

Legend

Complete

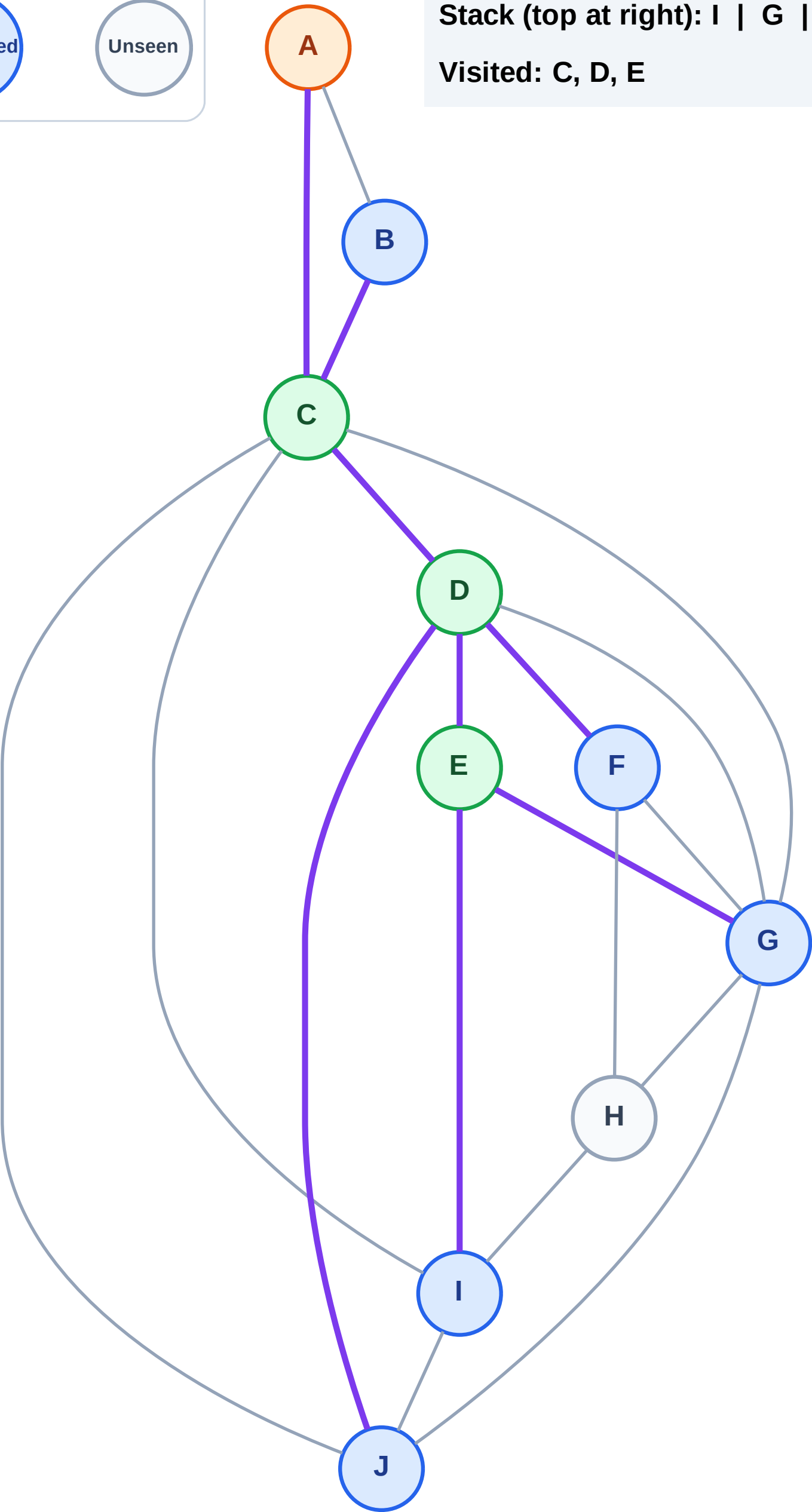
Processing

Discovered

Unseen

Stack (top at right): I | G | J | F | B

Visited: C, D, E



DFS Traversal

Step 9: No new neighbors from A

Legend

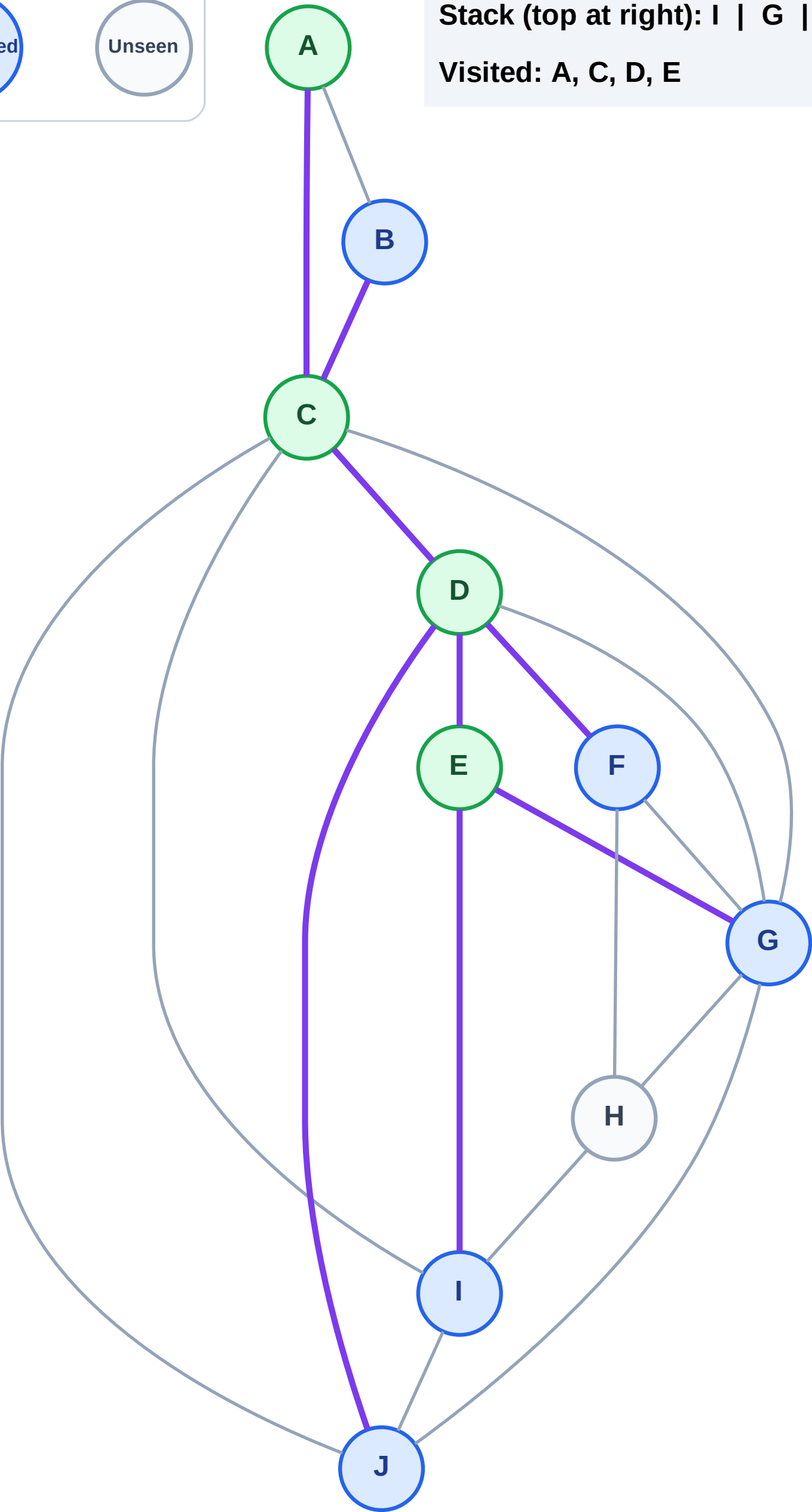
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | J | F | B
Visited: A, C, D, E



DFS Traversal

Step 10: Process node B

Legend

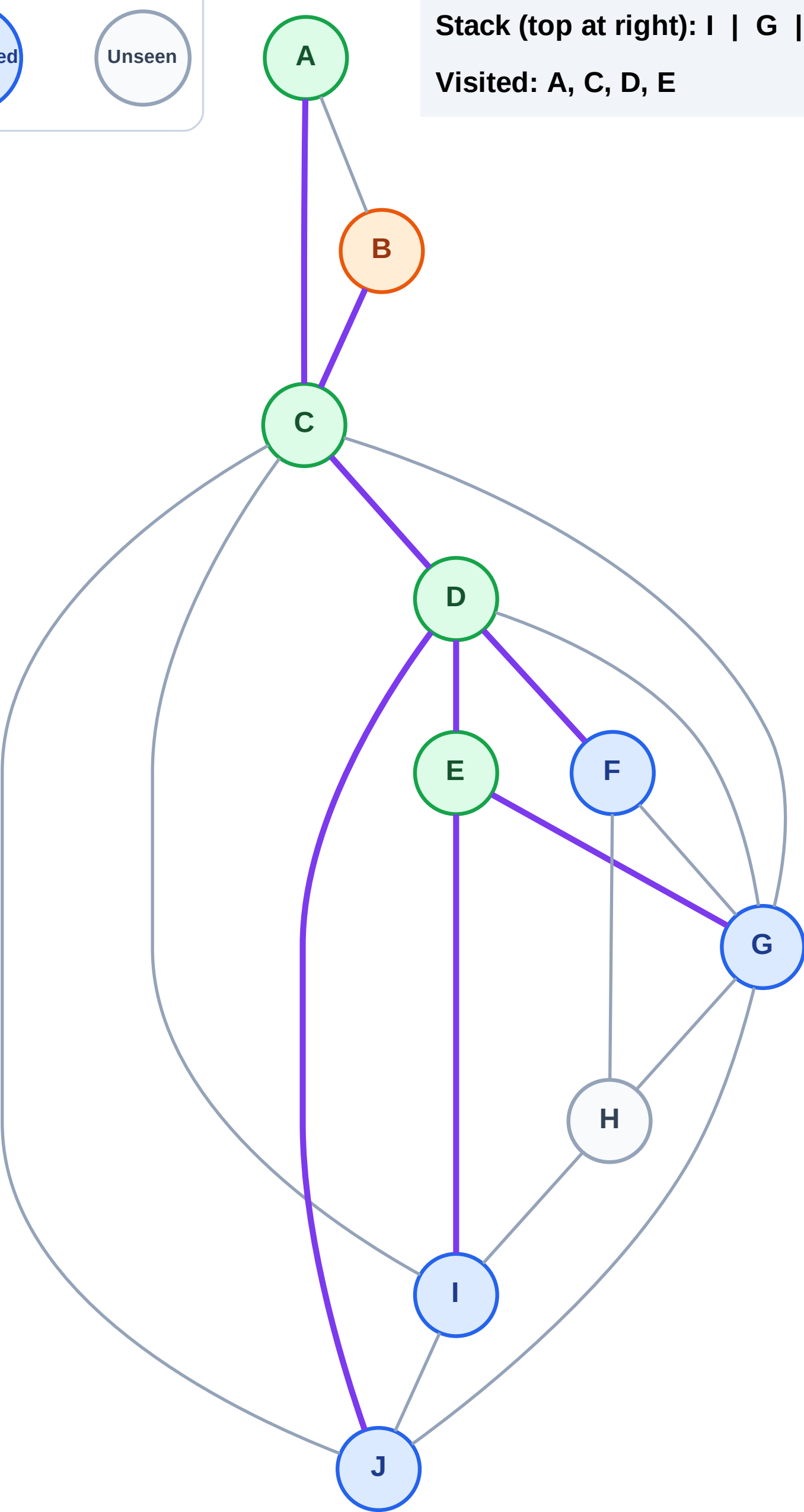
Complete

Processing

Discovered

Unseen

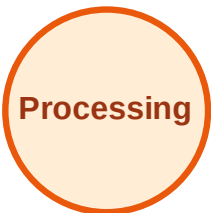
Stack (top at right): I | G | J | F
Visited: A, C, D, E



DFS Traversal

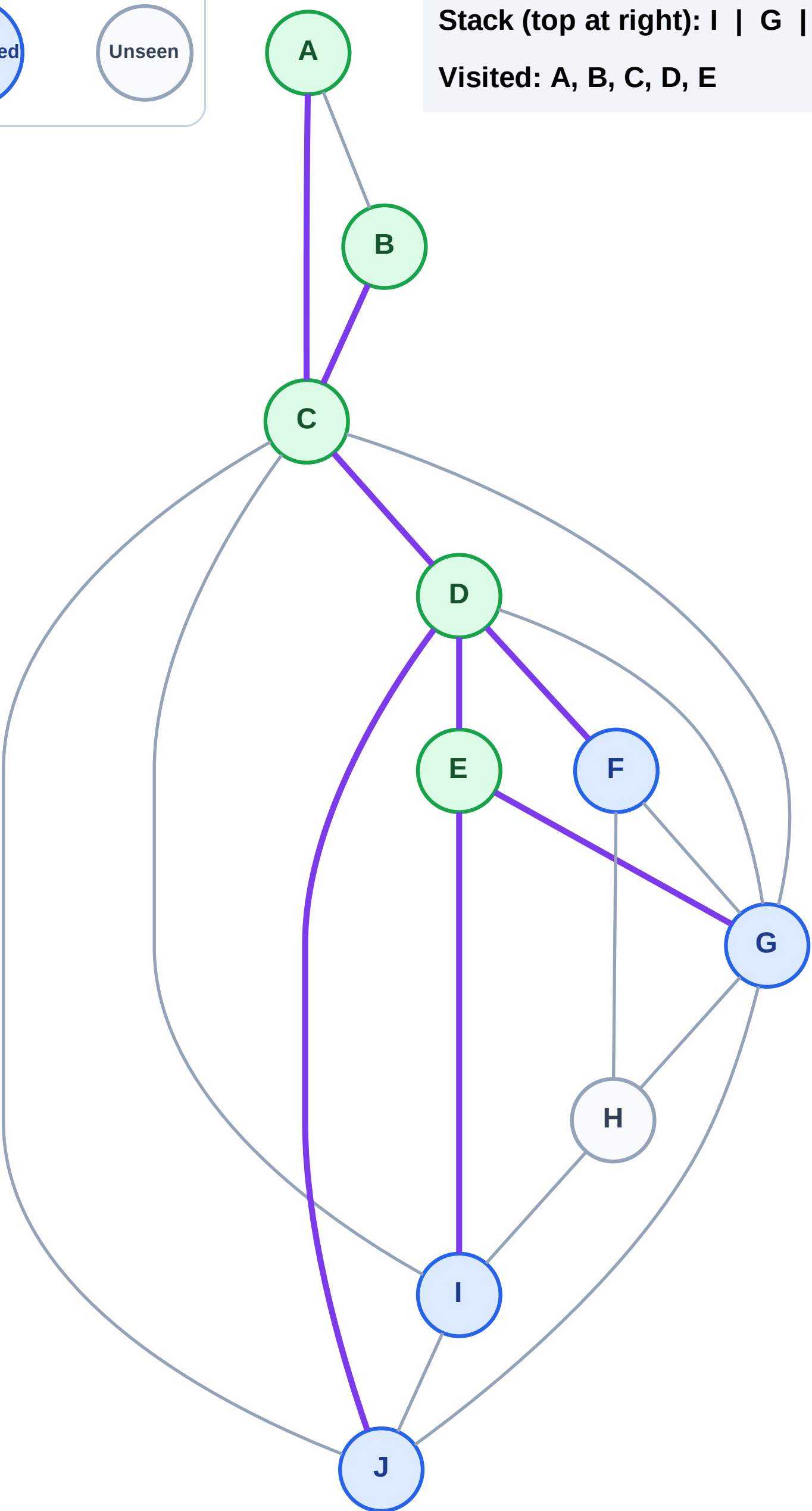
Step 11: No new neighbors from B

Legend



Stack (top at right): I | G | J | F

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node F

Legend

Complete

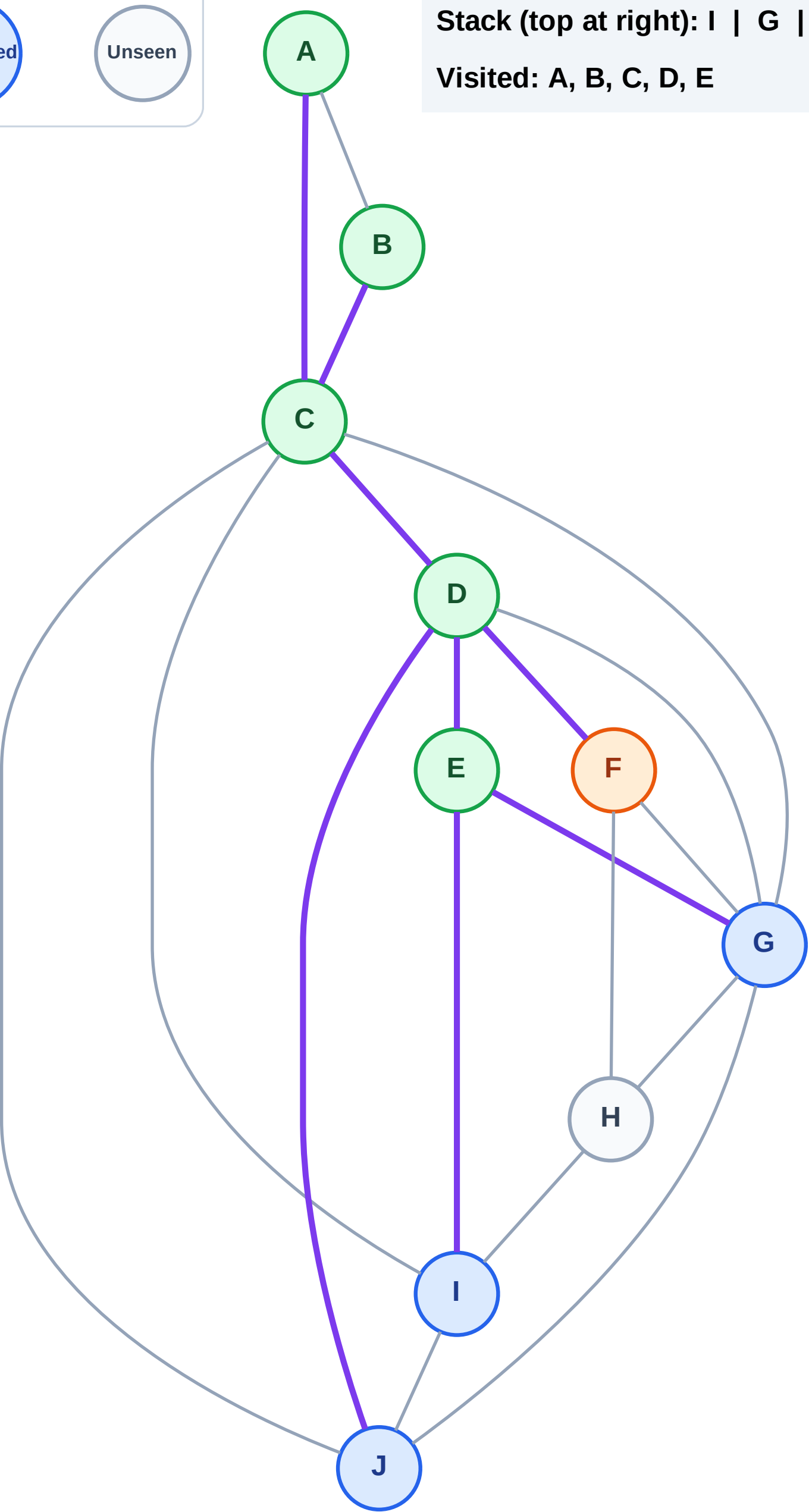
Processing

Discovered

Unseen

Stack (top at right): I | G | J

Visited: A, B, C, D, E



DFS Traversal

Step 13: Discover H from F

Legend

Complete

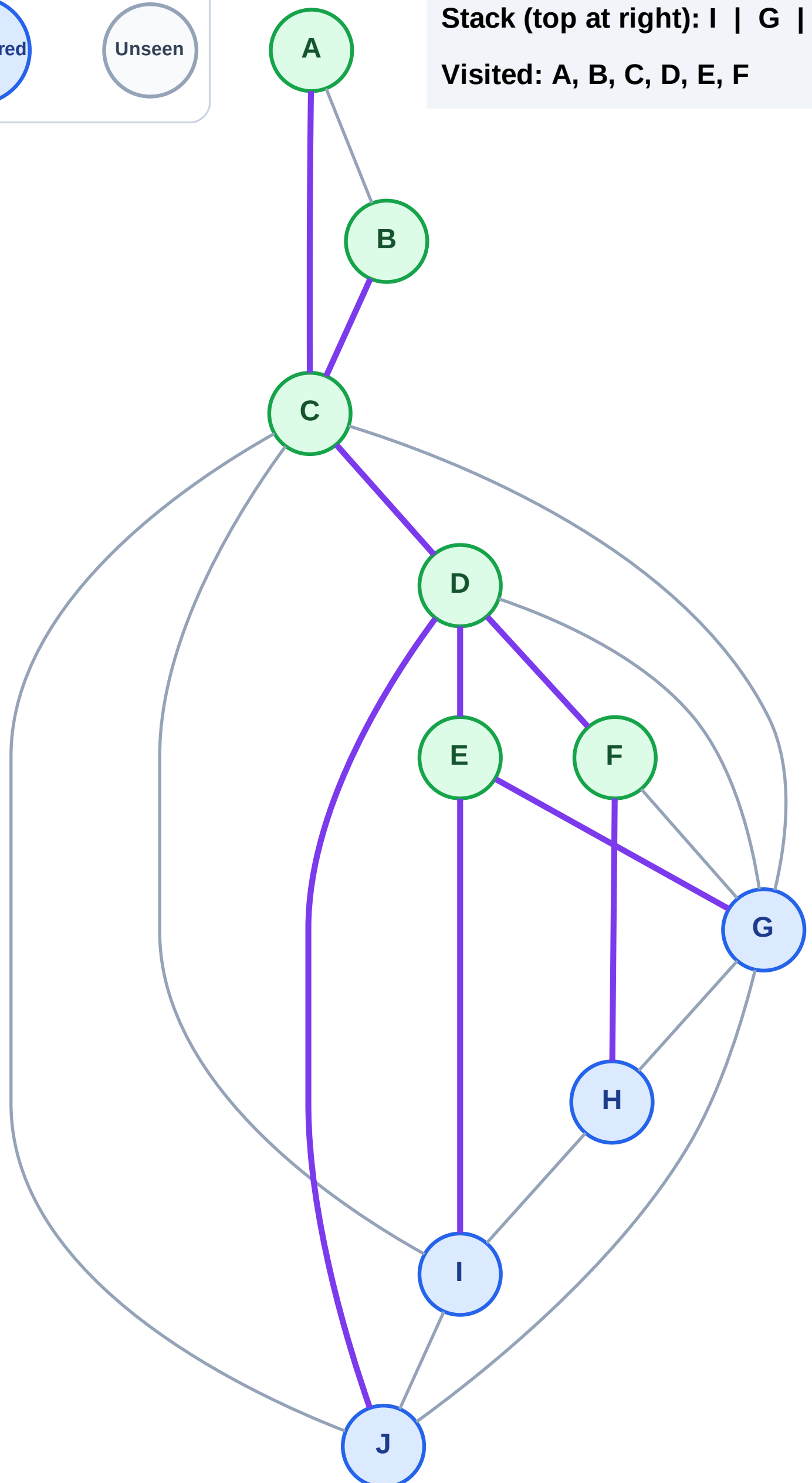
Processing

Discovered

Unseen

Stack (top at right): I | G | J | H

Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node H

Legend

Complete

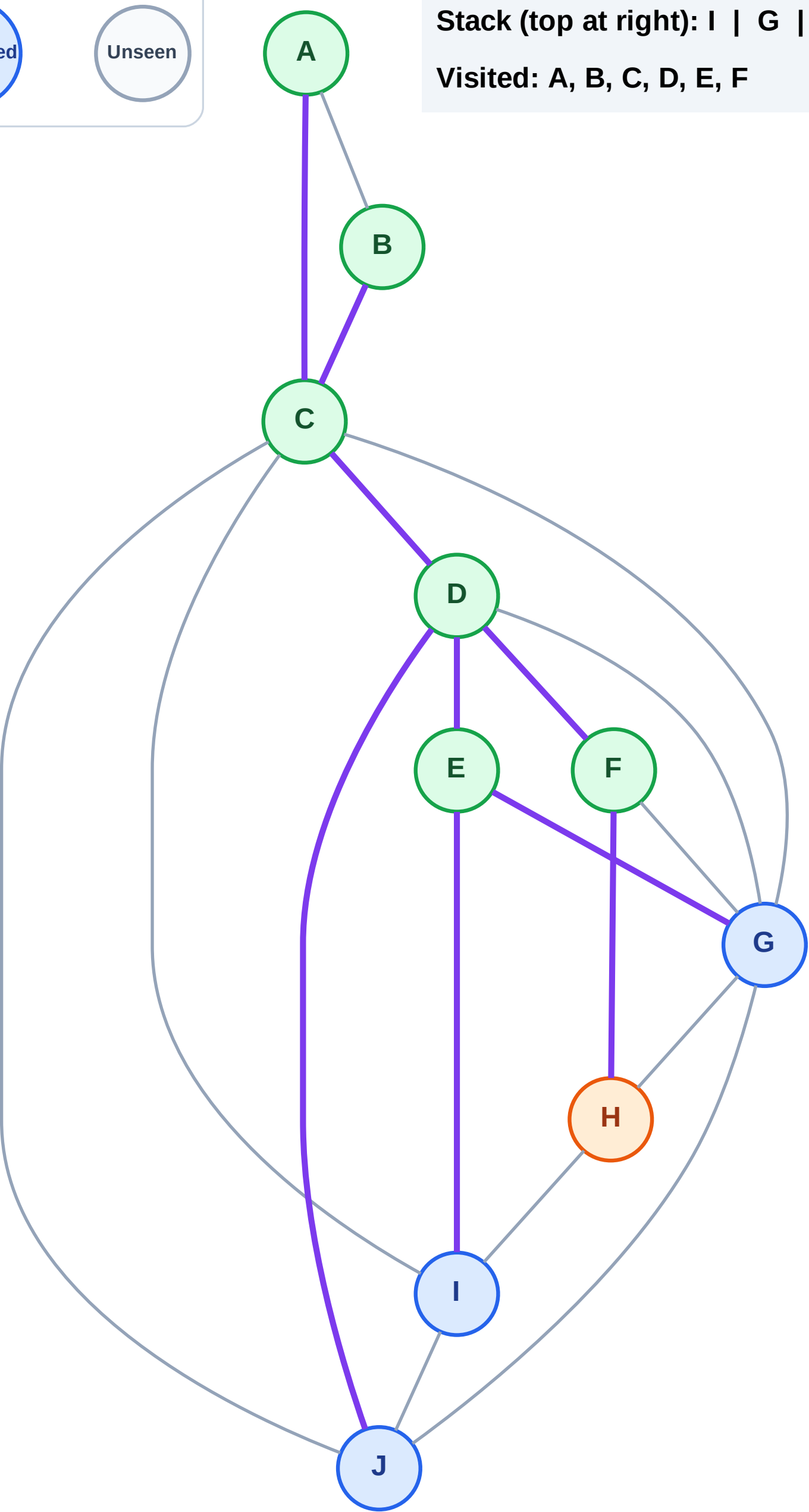
Processing

Discovered

Unseen

Stack (top at right): I | G | J

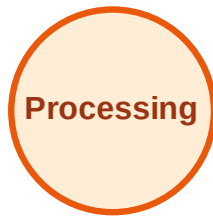
Visited: A, B, C, D, E, F



DFS Traversal

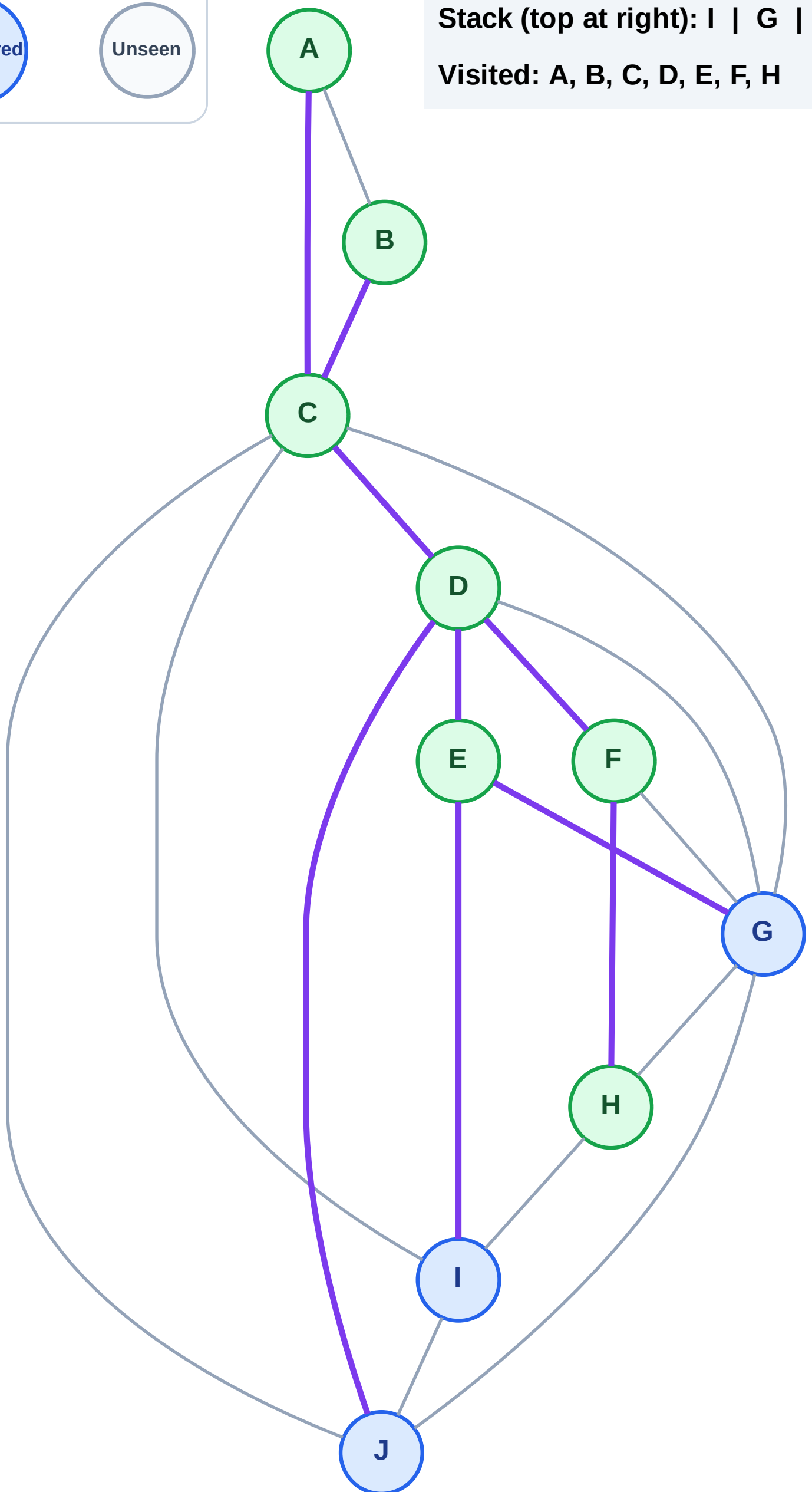
Step 15: No new neighbors from H

Legend



Stack (top at right): I | G | J

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node J

Legend

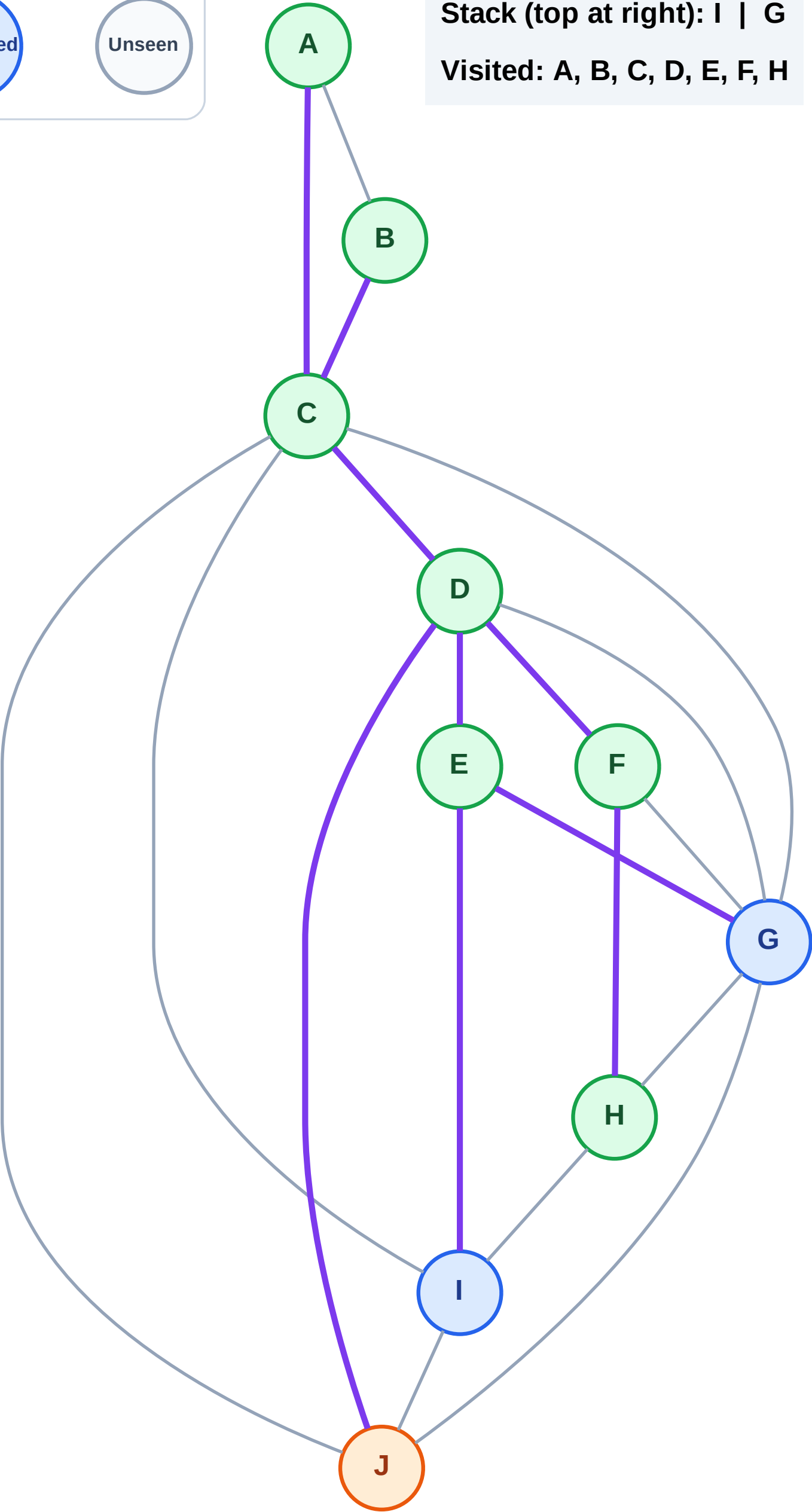
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G
Visited: A, B, C, D, E, F, H



DFS Traversal

Step 17: No new neighbors from J

Legend

Complete

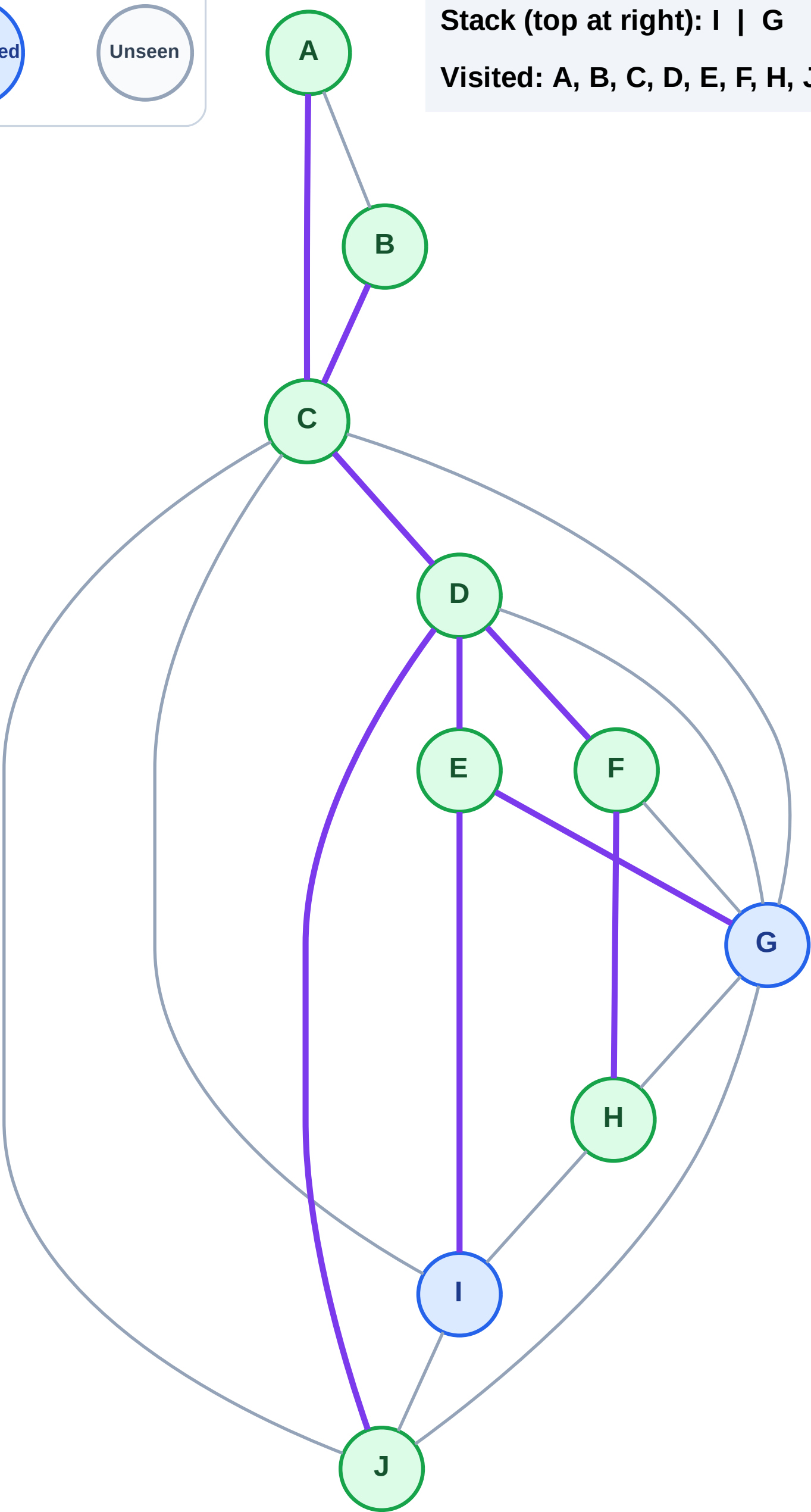
Processing

Discovered

Unseen

Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

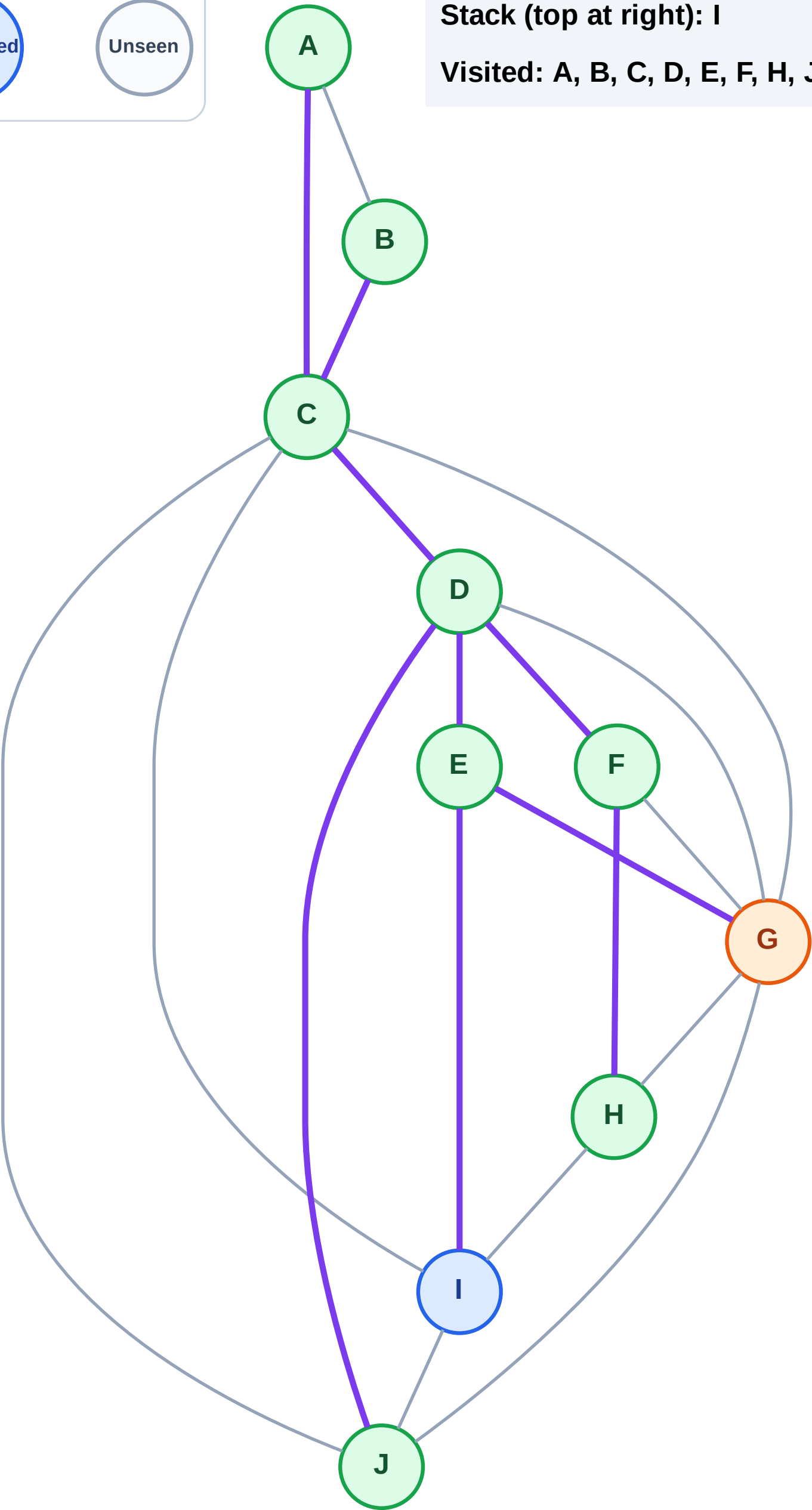
Complete

Processing

Discovered

Unseen

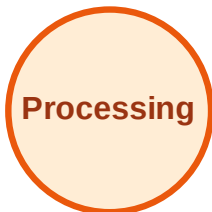
Stack (top at right): I
Visited: A, B, C, D, E, F, H, J



DFS Traversal

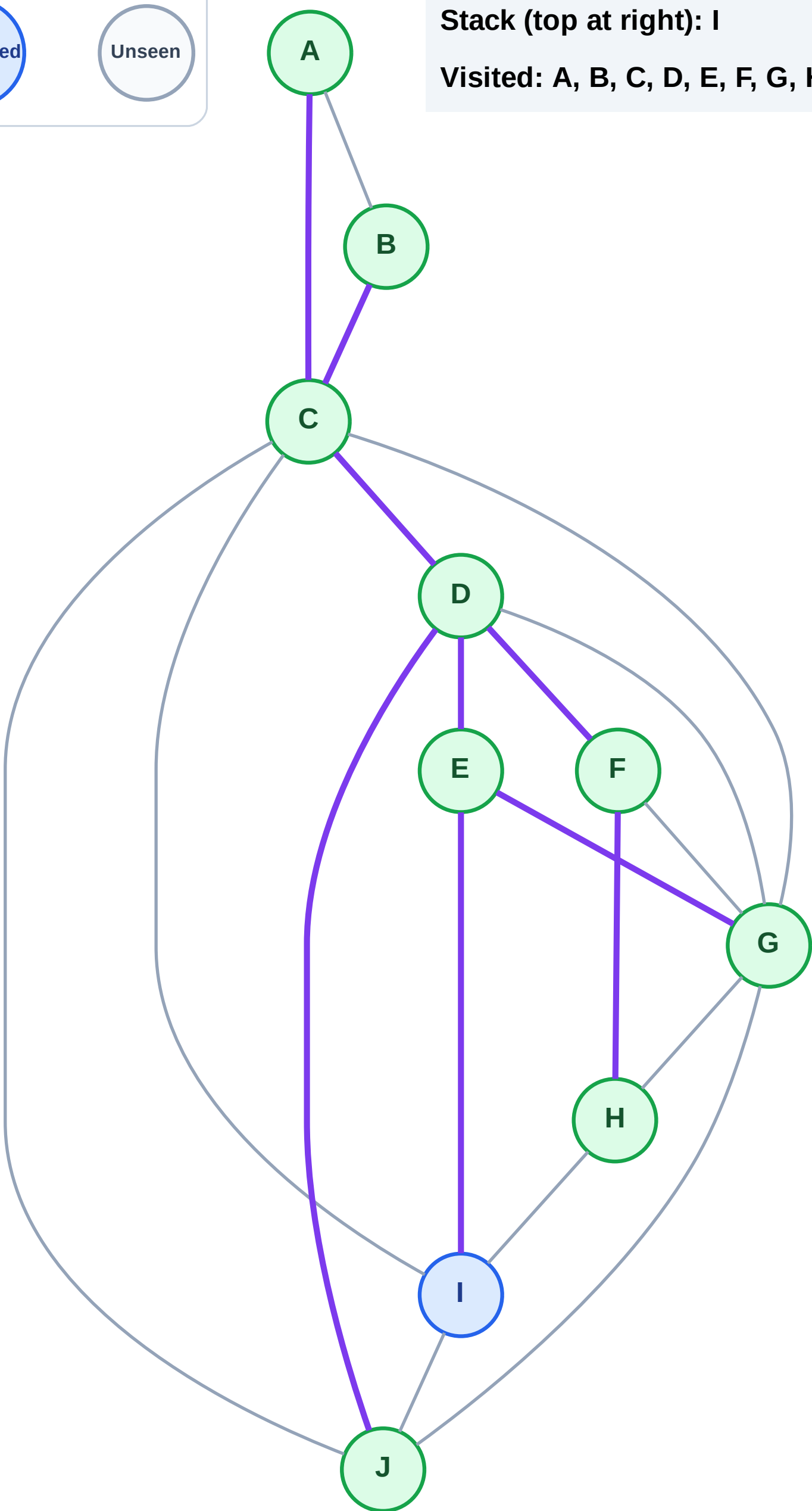
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

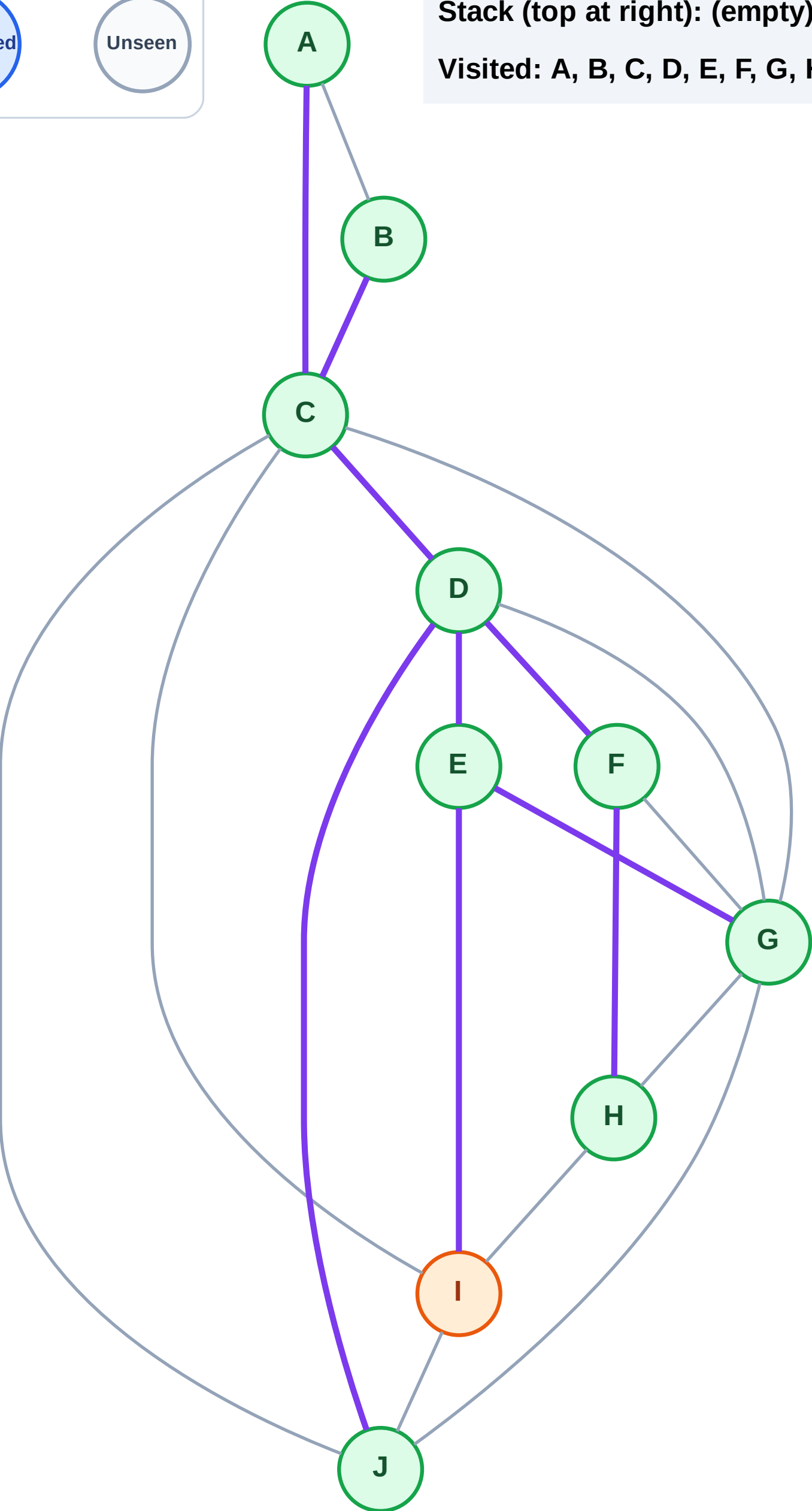
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

